# NEGATIVE-DEGREE HILBERT FUNCTIONS OF LAGRANGIAN AND ORTHOGONAL SCHUBERT VARIETIES

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ABSTRACT. Let $X_\kappa$ be the Schubert variety of dimension $|\kappa|$ in the Lagrangian Grassmannian $\mathrm{LG}(m, 2m)$, indexed by a strict partition $\kappa$, and let $H_\kappa(j) = \chi(X_\kappa, \mathcal{O}(j))$ be its Hilbert polynomial. For every $j \geq 1$, the number $(-1)^{|\kappa|} H_\kappa(-j)$ counts the strictly order-preserving maps from the shifted diagram of $\kappa$ to $\{1, \ldots, 2j-1\}$ that take even values on diagonal boxes. Equivalently, $H_\kappa$ is the Ehrhart polynomial of the order polytope of the shifted diagram with respect to the lattice that is $\mathbb{Z}$ on diagonal coordinates and $\frac{1}{2}\mathbb{Z}$ off the diagonal. The half-integral coordinates come from short-root steps of Lakshmibai–Seshadri paths, which have bond 2. Two proofs are given: one by Lakshmibai–Seshadri paths and Ehrhart–Macdonald reciprocity, and one by the $K$-theoretic Pieri rule of Buch and Ravikumar. As consequences, the integer zeros of $H_\kappa$ are exactly $-1, \ldots, -\lceil r/2 \rceil$, where $r$ is the length of a longest chain in the diagram, and the sign beyond them is $(-1)^{|\kappa|}$. The half-integer zeros and signs are determined completely, and $H_\kappa(-M/2) \neq 0$ for every odd $M \geq 2m+1$. For orthogonal Grassmannians $\mathrm{OG}(m, 2m+1)$ the analogous statements follow classically from minuscule standard monomial theory; $K$-theoretic proofs are included.

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## 1. Introduction

Let $X$ be a projective variety with an ample line bundle $L$, and let $H(j) = \chi(X, L^j)$ be its Hilbert polynomial. The values of $H$ at negative $j$ are where duality lives. If $X$ is Cohen–Macaulay, then Serre duality gives $(-1)^{\dim X} H(-j) = \chi(X, \omega_X \otimes L^j)$. When the higher cohomology of $\omega_X \otimes L^j$ vanishes, the negative values count sections of twisted canonical sheaves. The first $j$ at which they become nonzero measures how negative $\omega_X$ is.

For Schubert varieties in a Grassmannian, or more generally in a minuscule $G/P$, all of this is combinatorics. The homogeneous coordinate ring of a Schubert variety is an algebra with straightening law on a distributive lattice $J(P)$ and degenerates to the Hibi ring of that lattice [11, 10, 23]. Its Hilbert function is Stanley's order polynomial $j \mapsto \Omega(P, j+1)$. Stanley's reciprocity theorem $\Omega(P, -n) = (-1)^{|P|}\overline{\Omega}(P, n)$ [24, 26] then expresses the negative values as counts of strictly order-preserving maps; these index a basis of the canonical module of the Hibi ring.

The Lagrangian Grassmannian $\mathrm{LG}(m, 2m) = \mathrm{Sp}_{2m}/P_m$ is cominuscule but not minuscule, and the order-polynomial picture fails at once. Its Schubert varieties $X_\kappa$ are indexed by strict partitions $\kappa$ with parts at most $m$. The relevant poset is the shifted diagram $P_\kappa$ of $\kappa$, exactly as for the orthogonal Grassmannian $\mathrm{OG}(m, 2m+1)$, but the Hilbert functions are different. For $\kappa = (2,1)$, $X_\kappa = \mathrm{LG}(2,4)$ is a three-dimensional quadric with $H(j) = (j+1)(j+2)(2j+3)/6$. The order polynomial of the three-element chain $P_{(2,1)}$ is instead $\binom{j+3}{3}$, the Hilbert function of $\mathrm{OG}(2,5) \cong \mathbb{P}^3$. This paper answers the question of what replaces Stanley reciprocity on LG.

### 1.1. Main results.

For a strict partition $\kappa = (\kappa_1 > \cdots > \kappa_\ell > 0)$, the shifted diagram $D(\kappa)$ is the set of boxes $(a, b)$ with $0 \le a < \ell$ and $a \le b \le a + \kappa_{a+1} - 1$. We write $P_\kappa$ for $D(\kappa)$ with the product order ($(a,b) \le (a',b')$ iff $a \le a'$ and $b \le b'$). The boxes $(a,a)$ are the *diagonal boxes*. Let $r(\kappa)$ denote the number of elements of a longest chain in $P_\kappa$; by Lemma 2.2, $r(\kappa) = \max_i(\kappa_i + 2i - 2)$. For $\kappa$ with parts at most $m$, let $X_\kappa \subseteq \mathrm{LG}(m, 2m)$ be the Schubert variety of dimension $|\kappa|$ (§2.4), and put $H_\kappa(j) = \chi(X_\kappa, \mathcal{O}(j))$, where $\mathcal{O}(1)$ is the ample generator of $\mathrm{Pic}\,\mathrm{LG}(m, 2m)$.

**Theorem 1.1** (Strict reciprocity on LG)**.** *Let $\kappa$ be a strict partition with parts at most $m$.*

(a) *For every integer $j \geq 0$, $H_\kappa(j)$ is the number of order-preserving maps*

$$g\colon P_\kappa \to \{0, 1, \dots, 2j\} \quad \textit{with } g(x) \textit{ even for every diagonal box } x.$$

(b) *For every integer $j \geq 1$, $(-1)^{|\kappa|} H_\kappa(-j)$ is the number of strictly order-preserving maps*

$$f\colon P_\kappa \to \{1, \dots, 2j-1\} \quad \textit{with } f(x) \textit{ even for every diagonal box } x.$$

(c) *Equivalently, $H_\kappa(j) = \#(j\,\mathcal{O}(P_\kappa) \cap \Lambda)$ for all $j$. Here $\mathcal{O}(P_\kappa) \subseteq \mathbb{R}^{P_\kappa}$ is the order polytope and $\Lambda$ is the lattice of vectors whose diagonal coordinates lie in $\mathbb{Z}$ and whose other coordinates lie in $\frac{1}{2}\mathbb{Z}$. That is, $H_\kappa$ is the Ehrhart polynomial of $\mathcal{O}(P_\kappa)$ with respect to $\Lambda$.*

Part (a) is a short consequence of the Lakshmibai–Seshadri path model and may be known to experts (see §1.4). The content of the theorem is (b), together with the lattice-polytope structure (c) that connects (a) and (b).

**Example 1.2.** Take $\kappa = (2,1)$ and $m = 2$, so $X_\kappa = \mathrm{LG}(2,4)$. The poset $P_\kappa$ is the chain $(0,0) < (0,1) < (1,1)$, and both ends are diagonal. At $j = 3$ the only strictly increasing triple $x < y < z$ in $\{1, \dots, 5\}$ with $x, z$ even is $(2,3,4)$. At $j = 4$ there are 5 such triples, and at $j = 1, 2$ there are none. This matches $H(-1) = H(-2) = 0$, $H(-3) = -1$ and $H(-4) = -5$ for $H(j) = (j+1)(j+2)(2j+3)/6$. Example 4.11 returns to this shape: for $m = 3$ the poset $P_{(3)}$ is also a three-element chain, and only the marking of the diagonal separates it from $P_{(2,1)}$.

For the orthogonal Grassmannian $\mathrm{OG}(m, 2m+1)$, with Schubert varieties $X_\kappa^{\mathrm{OG}}$ of dimension $|\kappa|$ and the spinor line bundle as $\mathcal{O}(1)$, the analogue is classical.

**Theorem 1.3** (Strict reciprocity on OG)**.** *For every strict $\kappa$ with parts at most $m$, $H_\kappa^{\mathrm{OG}}(j) = \Omega(P_\kappa, j+1)$ for all $j$. For $j \geq 1$, $(-1)^{|\kappa|} H_\kappa^{\mathrm{OG}}(-j) = \overline{\Omega}(P_\kappa, j-1)$, the number of strictly order-preserving maps $P_\kappa \to \{1, \dots, j-1\}$.*

Theorem 1.3 is a composite of classical results: minuscule standard monomial theory and Stanley reciprocity (Remark 4.15). We also give a $K$-theoretic proof (§4.5) because it runs in parallel with the LG case.

The counts in Theorems 1.1 and 1.3 locate the integer zeros exactly.

**Theorem 1.4** (The integer window)**.** *Put $W(\kappa) = \lceil r(\kappa)/2 \rceil$ on* LG *and $W(\kappa) = r(\kappa)$ on* OG*, and $s_j(\kappa) = (-1)^{|\kappa|} H_\kappa(-j)$ (resp. with $H_\kappa^{\mathrm{OG}}$). Then*

$$0 = s_1(\kappa) = \cdots = s_W(\kappa) < s_{W+1}(\kappa) \leq s_{W+2}(\kappa) \leq \cdots.$$

*In particular, the zero set of $H_\kappa$ on the negative integers is exactly $\{-1, \dots, -W(\kappa)\}$, and $\operatorname{sgn} H_\kappa(-j) = (-1)^{|\kappa|}$ for $j > W(\kappa)$.*

Since $H_\kappa$ is a polynomial, it can be evaluated at half-integers. These values are not artefacts: $2^m H_\kappa(-a - \frac{1}{2}) = \chi(X_\kappa, \bigwedge^\bullet U \otimes \mathcal{O}(-a))$, where $U$ is the tautological subbundle (Proposition 6.3 and Remark 6.4). Their zeros obey the following law. Write $\kappa'$ for $\kappa$ with its largest part removed, and $N_\kappa(j)$ for the count in Theorem 1.1(b), with $N_\varnothing \equiv 1$.

**Theorem 1.5** (Half-integer values)**.** *Let $\kappa \neq \varnothing$. Then $H_\kappa(-\frac{1}{2}) = 2^{-\kappa_1}$, and for every integer $a \geq 1$:*

*(i) $H_\kappa(-a - \frac{1}{2}) = 0$ if and only if $\kappa' \neq \varnothing$ and $a \leq \lceil r(\kappa')/2 \rceil$;*
*(ii) otherwise, $\operatorname{sgn} H_\kappa(-a - \frac{1}{2}) = (-1)^{|\kappa| - \kappa_1}$ if $2a \leq \kappa_1$ and $(-1)^{|\kappa|}$ if $2a \geq \kappa_1$;*
*(iii) $|H_\kappa(-a - \frac{1}{2})| \geq 2^{-\kappa_1} N_{\kappa'}(a)$, with equality when $2a \leq \kappa_1$, and $2^{\kappa_1} H_\kappa(-a - \frac{1}{2}) \in \mathbb{Z}$.*

**Corollary 1.6.** *For every odd $M \geq 2m+1$, the number $(-1)^{|\kappa|} 2^m H_\kappa(-M/2)$ is a positive integer. In particular $H_\kappa$ has no zero at half-integers $\leq -m - \frac{1}{2}$.*

The vanishing half of Theorem 1.5(i) also has a separate proof (Theorem 6.7) by induction over hyperplane sections. That proof uses only the *support* of the $K$-theoretic Pieri rule, not its coefficients.

1.2. **Why the factor two.** The lattice $\Lambda$ has a simple origin. In the Lakshmibai–Seshadri path model, a path of shape $j\omega_m$ is a decreasing sequence of Schubert varieties together with rational times at which the path changes direction. A time $a$ is allowed between two consecutive directions exactly when $a\,\langle \tau(j\omega_m), \beta^\vee\rangle \in \mathbb{Z}$ for the roots $\beta$ along a Bruhat chain joining them. On LG, adding an off-diagonal box to a shifted diagram is a reflection in a short root $\varepsilon_i - \varepsilon_{i+1}$, and the relevant pairing (the *bond*) is 2. Adding a diagonal box is a reflection in the long root $2\varepsilon_m$, with bond 1 (Lemma 3.1). So off-diagonal boxes may enter at half-integral multiples of $1/j$, and diagonal boxes only at integral ones; this is the lattice $\Lambda$. Ehrhart–Macdonald reciprocity exchanges the closed polytope (weak inequalities, Theorem 1.1(a)) with its interior (strict inequalities, Theorem 1.1(b)). Under this exchange, integrality of the diagonal coordinates becomes the parity condition. On OG the weight $\omega_m$ is minuscule, every bond is 1, and one recovers Stanley's order polynomials.

The same dichotomy appears on the $K$-theoretic side. In Buch and Ravikumar's Pieri rule for LG [4], the boxes of a skew shape carry primed and unprimed letters, and diagonal boxes must be unprimed. In our recursion an odd value plays the role of a primed letter (Remark 4.9). We do not claim more than this dictionary. In particular, we have no general statement for other cominuscule, non-minuscule $G/P$ (Question 8.1).

1.3. **Method.** The paper contains two proofs of Theorem 1.1.

The first proof (§3) is elementary given two classical inputs: the Demazure character formula and Littelmann's path form of it [16, 17]. We compute the Bruhat covers on $W/W_{\omega_m}$ and their bonds. We then show that LS paths of shape $j\omega_m$ are in bijection with the maps in Theorem 1.1(a), and finish with Ehrhart–Macdonald reciprocity.

The second proof (§4) starts from a hyperplane section: $H_\kappa(j) - H_\kappa(j-1) = \sum_\rho c_\rho H_\rho(j)$, where the $c_\rho$ are the $K$-theoretic Pieri coefficients for multiplication by the Schubert divisor. We show (Theorem 4.8) that $c_\rho = (-1)^{|\theta|-1} V(\theta)$, where $\theta = D(\kappa) \setminus D(\rho)$ and $V(\theta)$ counts strictly order-preserving maps $\theta \to \{e < o\}$ that send diagonal boxes to $e$. The argument uses a recursion of Buch and Ravikumar that they prove in full [4, Thm. 5.5]. With this, the negative-degree recursion becomes the recursion that peels the top two values off a strict count, and no reciprocity theorem is needed. The second proof is longer, but it extends further: the same recursion, started at the half-integer anchor $H_\kappa(-\frac{1}{2}) = 2^{-\kappa_1}$, proves Theorem 1.5.

1.4. **Relation to the literature.** On the positive side, De Concini's symplectic standard tableaux [7] and Littelmann's LS paths [16] give bases of $H^0(X_\kappa, \mathcal{O}(j))$. Theorem 1.1(a) follows quickly from the latter.[1] Chirivì's semi-toric degenerations [5] and the recent work of Chirivì, Costa Cesari, Fang and Littelmann [6] relate LS paths on chains with prescribed bonds to lecture hall polytopes; their Gorenstein results concern interior LS paths on each toric component. We did not find in these works a statement about Schubert varieties in LG specifically, or about $\chi(X_\kappa, \mathcal{O}(-j))$. Stembridge's enriched $P$-partitions [27] satisfy a *self*-reciprocity, which is a different statement. For OG, Theorem 1.3 is classical (Remark 4.15). The $K$-theoretic Pieri rule we use is that of Buch and Ravikumar [4]. To the best of our knowledge, Theorem 1.1(b),

[1] R. Dehy [8], as reported by Mathieu [20, §4], equips $\{u \le w\}$ with a simplicial structure carrying an integral structure $P_w(n)$, and uses LS paths to interpret dimensions of Demazure modules as Ehrhart polynomials. Mathieu records that it was open whether $P_w$ can always be realised as a lattice polytope. We have not read [8] first-hand. As far as Mathieu's summary shows, it does not contain the lattice $\Lambda$ or Theorem 1.1(b), but it may anticipate the Ehrhart viewpoint of §3.

Theorem 4.8 as a closed formula, and the half-integer results (Theorem 1.5) have not appeared before.

1.5. **Outline.** Section 2 fixes notation and lists the cited inputs. Section 3 gives the LS-path proof of Theorem 1.1. Section 4 gives the $K$-theoretic proof, including the closed form of the Pieri coefficients and the OG case. Section 5 proves Theorem 1.4 in two ways. Section 6 treats half-integers: Theorem 1.5, Corollary 1.6, the hyperplane-section proof of the vanishing half, and an explicit half-step operator $T$ with $s_{x+1/2} = T\, s_x$ (Proposition 6.16). Section 7 collects examples, including a complete table for $\mathrm{LG}(3,6)$. Section 8 lists open questions.

## 2. Preliminaries

2.1. **Shifted diagrams.** Let $\Delta_m = \{(a,b) \in \mathbb{Z}^2 : 0 \le a \le b \le m-1\}$, ordered by the product order. We call $a$ the row and $b$ the column of $(a,b)$, and we draw row $a$ above row $a+1$. Then $x \le y$ means that $x$ is weakly north-west of $y$. For a strict partition $\kappa$ with $\kappa_1 \le m$, $D(\kappa) \subseteq \Delta_m$ is the shifted diagram defined in the introduction, and $P_\kappa = (D(\kappa), \le)$. The diagonal box $(a,a)$ is the leftmost box of row $a$. For $\rho \subseteq \kappa$ (meaning $\rho_i \le \kappa_i$ for all $i$), row $a$ of $D(\rho)$ is an initial segment of row $a$ of $D(\kappa)$.

**Lemma 2.1.** *The map $\kappa \mapsto D(\kappa)$ is a bijection from strict partitions with parts at most $m$ to order ideals of $\Delta_m$, and $\rho \subseteq \kappa$ iff $D(\rho) \subseteq D(\kappa)$. Consequently, for $\rho \subseteq \kappa$ the set $\theta = D(\kappa) \setminus D(\rho)$ is an up-set of $P_\kappa$, and every up-set of $P_\kappa$ arises in this way from a unique $\rho$.*

*Proof.* $D(\kappa)$ is clearly an order ideal. Conversely, let $I$ be an order ideal. If $(a,b) \in I$ and $b > a$, then $(a, b-1) \in I$, so each row of $I$ is an interval starting on the diagonal. If $(a,b) \in I$ with $a \ge 1$, then $(a-1, b) \in I$, so row $a-1$ reaches at least column $b$, that is, it is strictly longer than row $a$. Hence the row lengths form a strict partition. The remaining statements follow, since the complement of an up-set of $P_\kappa$ is an order ideal of $P_\kappa$, hence of $\Delta_m$. □

**Lemma 2.2.** *$r(\kappa) = 1 + \max\{a+b : (a,b) \in D(\kappa)\} = \max_{1 \le i \le \ell(\kappa)} (\kappa_i + 2i - 2)$, and $r(\varnothing) = 0$.*

*Proof.* Along a chain $x_1 < \cdots < x_r$ the coordinate sum strictly increases, so $r \le 1 + \max(a+b)$. Conversely, for $(a,b) \in D(\kappa)$ the staircase path $(0,0), (0,1), (1,1), (1,2), \ldots, (a,a), (a,a+1), \ldots, (a,b)$ consists of $a+b+1$ boxes. Each of them lies below $(a,b)$, so all of them lie in the order ideal $D(\kappa)$, and the path is a chain. The last box of row $i-1$ is $(i-1, i+\kappa_i-2)$, with coordinate sum $\kappa_i + 2i - 3$. □

For example, $r(4,3,2,1) = 7$ and $r(3,2) = 4$. Write $\kappa' = (\kappa_2, \kappa_3, \ldots)$. The map $(a,b) \mapsto (a-1, b-1)$ identifies the rows $\ge 1$ of $P_\kappa$ with $P_{\kappa'}$, preserving order and diagonal boxes.

2.2. **Counting functions.** For a finite poset $P$ and $n \ge 0$, let $\Omega(P,n)$ (resp. $\overline{\Omega}(P,n)$) be the number of order-preserving (resp. strictly order-preserving) maps $P \to [n] = \{1, \ldots, n\}$. For $D \subseteq P$, let

$$N(P, D; n) = \#\{f \colon P \to [n] \text{ strictly order-preserving} : f(D) \subseteq 2\mathbb{Z}\}.$$

For strict $\nu \subseteq \kappa$ and $j \ge 1$ put

$$N_{\kappa/\nu}(j) = N\big(D(\kappa) \setminus D(\nu),\ \text{diagonal boxes};\ 2j-1\big), \qquad N_\kappa(j) = N_{\kappa/\varnothing}(j),$$

so $N_\varnothing \equiv 1$. For $N \ge 0$ let $F_\kappa(N)$ be the number of order-preserving $g \colon P_\kappa \to \{0, \ldots, N\}$ that are even on diagonal boxes.

**Lemma 2.3.** *Let $\kappa \ne \varnothing$ and $r = r(\kappa)$.*

*(a) $N_\kappa(j) > 0$ iff $j \ge \lceil r/2 \rceil + 1$, and $N_\kappa(j) \le N_\kappa(j+1)$ for all $j \ge 1$.*
*(b) $\overline{\Omega}(P_\kappa, n) > 0$ iff $n \ge r$, and $\overline{\Omega}(P_\kappa, n) \le \overline{\Omega}(P_\kappa, n+1)$.*

*Proof.* (a) The map $f(a,b) = a+b+2$ is strictly order-preserving with values in $[2, r+1]$ (Lemma 2.2), and $f(a,a) = 2a+2$ is even. Conversely, along the staircase chain of Lemma 2.2, $f(0,0) \geq 2$ because $f(0,0)$ is even, and $f$ increases strictly over $r$ boxes, so its last value is at least $r+1$. Hence $N_\kappa(j) > 0$ iff $2j-1 \geq r+1$. Because $2j-1$ is odd, this is $j \geq r/2+1$ for $r$ even and $j \geq (r+3)/2$ for $r$ odd, that is, $j \geq \lceil r/2 \rceil + 1$. Monotonicity holds because $[2j-1] \subseteq [2j+1]$, and the inclusion preserves parity. (b) is the same argument with $f(a,b) = a+b+1$ and no parity constraint. □

2.3. **Reciprocity.** Let $Q \subseteq \mathbb{R}^d$ be a $d$-dimensional polytope whose vertices lie in a lattice $\Gamma \subseteq \mathbb{R}^d$ of rank $d$. Then $L_Q(t) = \#(tQ \cap \Gamma)$ is a polynomial in $t \in \mathbb{Z}_{\geq 0}$ of degree $d$. *Ehrhart–Macdonald reciprocity* [18, 2] says that $L_Q(-t) = (-1)^d \#(\mathrm{int}(tQ) \cap \Gamma)$ for $t \geq 1$. (For $\Gamma \neq \mathbb{Z}^d$, apply a linear isomorphism $\Gamma \cong \mathbb{Z}^d$.) The order polytope of a finite poset $P$ is

$$\mathcal{O}(P) = \{y \in [0,1]^P : x \leq x' \Rightarrow y_x \leq y_{x'}\}.$$

It is full-dimensional, and its vertices are the indicator vectors of the up-sets of $P$ [25]. Its interior is cut out by the strict versions of the facet inequalities, namely $0 < y_x < 1$ and $y_x < y_{x'}$ for $x < x'$. *Stanley's reciprocity* [24], [26, §3.15] says that $\Omega(P, -n) = (-1)^{|P|}\overline{\Omega}(P, n)$ for $n \geq 0$.[2]

2.4. **Schubert varieties.** *Type C.* Let $G = \mathrm{Sp}_{2m}$ with $\varepsilon$-coordinates, simple roots $\alpha_i = \varepsilon_i - \varepsilon_{i+1}$ $(i < m)$ and $\alpha_m = 2\varepsilon_m$, and $\omega = \omega_m = \varepsilon_1 + \cdots + \varepsilon_m$. Coroots are $(\varepsilon_i \pm \varepsilon_k)^\vee = \varepsilon_i \pm \varepsilon_k$ and $(2\varepsilon_i)^\vee = \varepsilon_i$. Let $P \supseteq B$ be the maximal parabolic subgroup with $W_P = \mathrm{Stab}_W(\omega) \cong S_m$. Then $G/P = \mathrm{LG}(m, 2m)$, and $L_\omega = \mathcal{O}(1)$ is the ample generator of the Picard group. The orbit $W\omega$ is $\{\pm 1\}^m \cong W/W_P$. For a sign vector $\mu$, let $w_\mu \in W^P$ be the minimal coset representative with $w_\mu \omega = \mu$, and let $X_\mu = \overline{Bw_\mu P/P}$, of dimension $\ell(w_\mu)$. Put $\kappa(\mu) = \{m - i + 1 : \mu_i = -1\}$, a strict partition with parts at most $m$. By Lemma 3.1 below, $\mu \mapsto \kappa(\mu)$ is a bijection and $\dim X_\mu = |\kappa(\mu)|$. We write $X_\kappa = X_\mu$ and $w_\kappa = w_\mu$ when $\kappa = \kappa(\mu)$. Thus $w_\kappa \omega = v_\kappa$, the sign vector with $-1$ exactly in positions $m+1-k$, $k \in \kappa$.

*Type B.* Let $G = \mathrm{Spin}_{2m+1}$ with $\alpha_m = \varepsilon_m$ (short), $\alpha_m^\vee = 2\varepsilon_m$, and $\omega_m = \frac{1}{2}(\varepsilon_1 + \cdots + \varepsilon_m)$. Then $G/P_m = \mathrm{OG}(m, 2m+1)$ and $L_{\omega_m}$ is the spinor line bundle, the ample generator. Schubert varieties $X_\kappa^{\mathrm{OG}}$ are indexed in the same way.

For either type, $H_\kappa(j) = \chi(X_\kappa, L^j)$ is a polynomial in $j$ of degree $|\kappa|$ [13], and $H_\varnothing \equiv 1$ because $X_\varnothing$ is a point.

*Buch and Ravikumar's indexing.* Buch and Ravikumar [4, §§4–5] index Schubert varieties by codimension. On $\mathrm{LG}(m, 2m)$ they fix the basis $e_1, \ldots, e_{2m}$ of $\mathbb{C}^{2m}$ with $(e_i, e_k) = \pm\delta_{i+k,2m+1}$ (sign $+$ for $i < k$), the flag $E_p = \langle e_1, \ldots, e_p \rangle$, and for a strict $\lambda$ with $\lambda_1 \leq m$ put

$$X^\lambda(E_\bullet) = \{V \in \mathrm{LG}(m, 2m) : \dim(V \cap E_{m+1-\lambda_i}) \geq i \text{ for } 1 \leq i \leq \ell(\lambda)\},$$

a Schubert variety of codimension $|\lambda|$, with $\mathcal{O}^\lambda = [\mathcal{O}_{X^\lambda(E_\bullet)}]$. On $\mathrm{OG}(m, 2m+1)$ the definition is the same, for the form $(e_i, e_k) = \delta_{i+k,2m+2}$ on $\mathbb{C}^{2m+1}$. For strict $\lambda$ with parts at most $m$ let $\lambda^\vee$ be the strict partition whose parts are $[m] \setminus \lambda$ (BR's $m$-dual). Thus $|\lambda| + |\lambda^\vee| = \binom{m+1}{2} = \dim X$.

**Lemma 2.4** (Dictionary)**.** *Realise* $\mathrm{Sp}_{2m}$ *as the isometry group of Buch and Ravikumar's form on* $\mathbb{C}^{2m}$*. Let* $T$ *be its diagonal torus* $t = \mathrm{diag}(t_1, \ldots, t_m, t_m^{-1}, \ldots, t_1^{-1})$*, with* $\varepsilon_i(t) = t_i$*, and let* $B$ *be its subgroup of upper-triangular matrices. Then:*

*(a)* $B$ *is a Borel subgroup whose positive roots are* $\varepsilon_i \pm \varepsilon_k$ *(*$i < k$*) and* $2\varepsilon_i$*, so the conventions above apply. Moreover* $P = \mathrm{Stab}_G(E_m)$*, and* $gP \mapsto gE_m$ *identifies* $G/P$ *with* $\mathrm{LG}(m, 2m)$*.*

[2] We cite [26] by section; the theorem number in the second edition was not checked first-hand.

*(b) For every strict $\kappa$ with parts at most $m$,*

$$X_\kappa = X^{\kappa^\vee}(E_\bullet) \qquad \text{as subvarieties of } \mathrm{LG}(m,2m).$$

*In particular $[\mathcal{O}_{X_\kappa}] = \mathcal{O}^{\kappa^\vee}$ in $K(X)$, and BR's divisor class $\mathcal{O}^1$ is the class $[\mathcal{O}_{X^{(1)}}]$ of (G0) below.*

*The same holds on $\mathrm{OG}(m,2m+1)$, with $T = \{\mathrm{diag}(t_1,\dots,t_m,1,t_m^{-1},\dots,t_1^{-1})\}$ in $\mathrm{SO}_{2m+1}$ (on which $\mathrm{Spin}_{2m+1}$ acts through $\mathrm{SO}_{2m+1}$) and $B$ the upper-triangular subgroup.*

*Proof.* (a) The basis vector $e_i$ has $T$-weight $\varepsilon_i$ for $i \le m$, and $e_{2m+1-i}$ has weight $-\varepsilon_i$. The matrix entry in position $(p,q)$ has weight $\mathrm{wt}(e_p) - \mathrm{wt}(e_q)$. For $p<q$ these weights are $\varepsilon_i - \varepsilon_k$ ($p=i<q=k\le m$), $\varepsilon_i+\varepsilon_k$ ($p=i$, $q=2m+1-k$, $i\ne k$) and $2\varepsilon_i$ ($p=i$, $q=2m+1-i$). This is the positive system of the type-$C$ paragraph, with simple roots $\varepsilon_i-\varepsilon_{i+1}$ and $2\varepsilon_m$. Upper-triangular matrices preserve $E_m$, and $E_m$ is a Lagrangian, so $\mathrm{Stab}_G(E_m)$ is a parabolic subgroup containing $B$. Its Levi factor $\mathrm{GL}(E_m)$ has Weyl group $S_m = \mathrm{Stab}_W(\omega)$, so it is $P$. Since $G$ acts transitively on Lagrangians (Witt), $G/P \cong \mathrm{LG}(m,2m)$.

(b) Let $\mu = \mu(\kappa)$ be the sign vector with $\kappa(\mu)=\kappa$, and $w = w_\mu$. A representative $\dot w \in N_G(T)$ maps the weight space of a weight $\chi$ to that of $w\chi$. Since $w\omega = \mu$, the weights $w\varepsilon_1,\dots,w\varepsilon_m$ are $\mu_1\varepsilon_1,\dots,\mu_m\varepsilon_m$, and so

$$wP \;\longleftrightarrow\; V_\mu := \dot w E_m = \langle f_1,\dots,f_m\rangle, \qquad f_i = e_i \ (\mu_i=+1), \quad f_i = e_{2m+1-i} \ (\mu_i=-1).$$

Hence $X_\kappa = \overline{B\cdot V_\mu}$. Put $\lambda = \kappa^\vee$. By definition $\mu_i = -1$ iff $m+1-i \in \kappa$, so $\mu_i = +1$ iff $m+1-i \in \lambda$. For $p \le m$,

$$\dim(V_\mu \cap E_p) = \#\{i \le p : \mu_i = +1\} = \#\{\text{parts of } \lambda \text{ that are } \ge m+1-p\}.$$

At $p = m+1-\lambda_k$ this is $\#\{\text{parts} \ge \lambda_k\} = k$, so $V_\mu \in X^\lambda(E_\bullet)$. The variety $X^\lambda(E_\bullet)$ is closed and stable under $B$, which fixes every $E_p$. So $X_\kappa \subseteq X^\lambda(E_\bullet)$. Both are irreducible, and $\dim X_\kappa = |\kappa|$ (Lemma 3.1(a)), while $\dim X^\lambda(E_\bullet) = \binom{m+1}{2} - |\lambda| = |\kappa|$ [4, §§4–5]. Hence they are equal. Finally, $X^{(1)}(E_\bullet) = \{V : V \cap E_m \ne 0\}$ is the unique Schubert divisor.

In type $B$ the weights of $e_i$, $e_{m+1}$ and $e_{2m+2-i}$ are $\varepsilon_i$, $0$ and $-\varepsilon_i$. The same computation gives the positive roots $\varepsilon_i \pm \varepsilon_k$ and $\varepsilon_i$, and $P = \mathrm{Stab}(E_m)$. A $T$-stable isotropic $m$-space cannot contain $e_{m+1}$, because $(e_{m+1}, e_{m+1}) = 1$, so the $T$-fixed points are again the $V_\mu$. The dimension count of Lemma 3.1(a) is unchanged, because the coroots $\varepsilon_i \pm \varepsilon_k$ and $2\varepsilon_i$ pair with $\mu/2$ with the same signs as the type-$C$ coroots pair with $\mu$. The rest of the argument is identical. □

All Borel subgroups and maximal tori of $G$ are conjugate, and translation by $g \in G$ preserves $\mathcal{O}(1)$, Hilbert polynomials and $K$-classes. So we may and do use the coordinates of Lemma 2.4 throughout. The poset of shifted diagrams alone would not fix the dictionary: for $m=3$ it has an automorphism exchanging (3) and (2,1), and their Hilbert polynomials differ (Table 1). Lemma 2.4 gives a geometric check: $X_{(2,1)} = X^{(3)}(E_\bullet) = \{V : e_1 \in V\} \cong \mathrm{LG}(2, e_1^\perp/\langle e_1\rangle)$, the three-dimensional quadric, as in Example 1.2. Example 4.11 below gives a check in $K$-theory.

2.5. **Demazure operators and cited inputs.** For a simple root $\alpha_i$, the Demazure operator on $\mathbb{Z}[\text{weights}]$ is $D_i f = (f - e^{-\alpha_i} s_i f)/(1-e^{-\alpha_i})$. For $n = \langle \lambda, \alpha_i^\vee\rangle$,

$$D_i e^\lambda = \begin{cases} e^\lambda + e^{\lambda-\alpha_i} + \cdots + e^{s_i\lambda}, & n \ge 0,\\ 0, & n = -1,\\ -(e^{\lambda+\alpha_i} + \cdots + e^{s_i\lambda-\alpha_i}), & n \le -2. \end{cases}$$

For a reduced word $w = s_{i_1}\cdots s_{i_k}$, put $D_w = D_{i_1}\circ\cdots\circ D_{i_k}$; it does not depend on the reduced word. Littelmann's operator $\Lambda_\alpha(e^\mu) = (e^{\mu+\rho} - e^{s_\alpha(\mu+\rho)})/(1-e^{-\alpha})\cdot e^{-\rho}$ equals $D_\alpha$, because $s_\alpha(\mu+\rho) = s_\alpha\mu + \rho - \alpha$ for simple $\alpha$.

The following inputs are standard. We cite them and do not reprove them.

(G0) (Chevalley.) $\operatorname{Pic} X = \mathbb{Z}L$, and the unique Schubert divisor $X^{(1)}$ is the reduced zero scheme of a section of $L$. Hence $\mathcal{O}^1 := [\mathcal{O}_{X^{(1)}}] = 1 - [L^{-1}]$ in $K(X)$.

(G1) (Demazure character formula [9, 12, 1, 22, 14, 19]; see [3, Ch. 3].) For $w \in W^P$ and dominant $\lambda$ with $W_P \subseteq \operatorname{Stab}_W(\lambda)$, $H^i(X_w, L_\lambda) = 0$ for $i > 0$, and $\operatorname{char} H^0(X_w, L_\lambda)^* = D_w(e^\lambda)$.

(LS) (Littelmann [16, §5.2]; see [17, §8, p. 305].) For dominant $\lambda$ and $w \in W/W_\lambda$, $D_w(e^\lambda) = \sum_\pi e^{\pi(1)}$, where the sum runs over LS paths $\pi$ of shape $\lambda$ with initial direction $\leq w$. This holds for every symmetrizable Kac–Moody algebra, in particular in type $C_m$.[3]

(BR) (Buch–Ravikumar [4].) The $K$-theoretic Pieri rule for $\mathrm{LG}(m, 2m)$ and $\mathrm{OG}(m, 2m+1)$; the precise statements used are quoted in §4.2.

Combining (G1) with (LS) gives $h^0(X_\kappa, \mathcal{O}(j)) = \#\{\text{LS paths of shape } j\omega \text{ with initial direction} \leq w_\kappa\}$. Only this count is used in §3.

The next lemma lets us evaluate Demazure expressions at non-integral weights. Work with analytic germs at $h = 0$ on $\mathfrak{h} = \mathbb{C}^m$, with $e^\mu(h) = e^{\langle \mu, h\rangle}$ for $\mu \in \mathbb{C}^m$. Each $D_i$ maps germs to germs, because the numerator vanishes on $\alpha_i(h) = 0$ and $1 - e^{-\alpha_i}$ is $\alpha_i$ times a unit.

**Lemma 2.5** (Polynomial extension)**.** *For $w \in W$, $P_w(\lambda) := [D_w e^\lambda](0)$ is a polynomial in $\lambda \in \mathbb{C}^m$ of degree at most $\ell(w)$. Given (G1), $H_\kappa(j) = P_{w_\kappa}(j\omega)$ for every $j \in \mathbb{C}$.*

*Proof.* Let $\mathfrak{m}$ be the maximal ideal of germs at 0. For $f \in \mathfrak{m}^k$ with $k \geq 1$,

$$D_i f = \frac{f - s_i f}{1 - e^{-\alpha_i}} + s_i f,$$

and $(f - s_i f)/\alpha_i \in \mathfrak{m}^{k-1}$. Hence $D_i(\mathfrak{m}^k) \subseteq \mathfrak{m}^{k-1}$ and $D_w(\mathfrak{m}^{\ell(w)+1}) \subseteq \mathfrak{m}$. Expanding $e^\lambda = \sum_{k \leq \ell(w)} \langle \lambda, h\rangle^k / k! + R$ with $R \in \mathfrak{m}^{\ell(w)+1}$ shows that $P_w(\lambda)$ is a polynomial in $\lambda$. For integral $\lambda$, $D_w e^\lambda$ is a Laurent polynomial, and its value at $h = 0$ is its coefficient sum. By (G1), $H_\kappa(j) = P_{w_\kappa}(j\omega)$ for integers $j \geq 0$, and both sides are polynomials in $j$. □

**Lemma 2.6** (The anchor)**.** *If $\kappa \neq \varnothing$, then $H_\kappa(-1) = 0$, both on* LG *and on* OG.

*Proof.* Let $w = w_\kappa$. Since $w$ is minimal in $wW_P$, $ws_i > w$ for every $i \neq m$. Since $w \neq e$, it has a right descent, so $ws_m < w$, and we may write $w = w's_m$ with lengths adding. Then $D_w = D_{w'} D_m$, and $D_m$ acts first. In both types $\langle -\omega_m, \alpha_m^\vee \rangle = -1$, so $D_m e^{-\omega_m} = 0$. By Lemma 2.5, $H_\kappa(-1) = P_w(-\omega_m) = 0$. □

## 3. The LS-path proof of strict reciprocity

This section proves Theorem 1.1 using only (G1) and (LS). It uses neither the Buch–Ravikumar Pieri rule nor (G0).

3.1. **LS paths.** We recall Littelmann's definition [16, 17]. Let $\lambda$ be dominant, and give $W^\lambda = W/W_\lambda$ the Bruhat order. For $\sigma > \tau$ in $W^\lambda$ and a rational $a \in (0, 1)$, an *$a$-chain* for $(\sigma, \tau)$ is a sequence of Bruhat covers $\sigma = \kappa_0 > \kappa_1 > \cdots > \kappa_r = \tau$ with $\kappa_{k-1} = s_{\beta_k} \kappa_k$ ($\beta_k > 0$) and $a\langle \kappa_k(\lambda), \beta_k^\vee \rangle \in \mathbb{Z}$ for all $k$. An *LS path of shape $\lambda$* is a pair

$$\pi = (\tau_1 > \tau_2 > \cdots > \tau_s \ ; \ 0 = a_0 < a_1 < \cdots < a_s = 1)$$

such that an $a_i$-chain for $(\tau_i, \tau_{i+1})$ exists for $1 \leq i \leq s-1$. Its *initial direction* is $\tau_1$, and $\pi(1) = \sum_i (a_i - a_{i-1}) \tau_i(\lambda)$. We call $|\langle \kappa_k(\lambda), \beta_k^\vee \rangle|$ the *bond* of the cover $\kappa_{k-1} > \kappa_k$. The absolute value makes it independent of sign conventions, because $\langle s_\beta \mu, \beta^\vee \rangle = -\langle \mu, \beta^\vee \rangle$.

[3] We read the statement first-hand in the ICM survey [17, p. 305]. The locator §5.2 in [16] is quoted from Chirivì, Fang and Littelmann (arXiv:2207.08904) and was not read first-hand.

### 3.2. **Covers and bonds.**

**Lemma 3.1.** *In type $C_m$ with $\lambda = \omega_m$:*

*(a)* $\dim X_\mu = \#\{\alpha > 0 : \langle \mu, \alpha^\vee \rangle < 0\} = \sum_{i:\mu_i=-1}(m-i+1) = |\kappa(\mu)|$;

*(b) the Bruhat covers $\mu < s_\beta \mu$ in $W^\lambda$ are exactly the moves for which $D(\kappa(s_\beta\mu))$ is $D(\kappa(\mu))$ plus one box;*

*(c) the bond of such a cover is 1 if the added box is diagonal and 2 otherwise.*

*In particular, the Bruhat order on $W^\lambda$ is inclusion of shifted diagrams.*

*Proof.* (a) Let $w = w_\mu$. Since $w$ is minimal in $wW_P$, $w$ maps the positive roots of $W_P$ to positive roots. So if $\alpha > 0$ and $w^{-1}\alpha < 0$, then $w^{-1}\alpha$ is not a root of $W_P$, and $\langle \mu, \alpha^\vee \rangle = \langle \omega, (w^{-1}\alpha)^\vee \rangle < 0$. Conversely $\langle \mu, \alpha^\vee \rangle < 0$ forces $w^{-1}\alpha < 0$, since $\omega$ is dominant. Hence $\dim X_\mu = \ell(w) = \#\{\alpha > 0 : w^{-1}\alpha < 0\} = \#\{\alpha > 0 : \langle \mu, \alpha^\vee \rangle < 0\}$. The positive roots are $\varepsilon_i \pm \varepsilon_k$ $(i < k)$ and $2\varepsilon_i$. The pairing $\langle \mu, \varepsilon_i - \varepsilon_k \rangle$ is negative iff $\mu_i = -1$ and $\mu_k = +1$; $\langle \mu, \varepsilon_i + \varepsilon_k \rangle < 0$ iff $\mu_i = \mu_k = -1$; and $\langle \mu, \varepsilon_i \rangle < 0$ iff $\mu_i = -1$. So each $i$ with $\mu_i = -1$ contributes $1 + (m - i)$.

(b, c) Let $i < k$, $p = m - i + 1$ and $q = m - k + 1$, so $p > q$. The reflection $s_{\varepsilon_i - \varepsilon_k}$ swaps $\mu_i$ and $\mu_k$. Upward $(\mu_i = +1, \mu_k = -1)$ it replaces the part $q$ by $p$ and raises the dimension by $p - q$. It is a cover iff $p = q + 1$, that is, iff it adds one box at the end of the row of length $q$. That box is off the diagonal because $q \geq 1$, and the bond is $|1 - (-1)| = 2$. The reflection $s_{\varepsilon_i + \varepsilon_k}$ flips $\mu_i, \mu_k$ when they are equal. Upward it adds the two parts $p$ and $q$, raising the dimension by $p + q \geq 3$, so it is never a cover. The reflection $s_{2\varepsilon_i}$ flips $\mu_i$. Upward it adds the part $p$, so it is a cover iff $p = 1$ (that is, $i = m$): a new row of length one, whose box is diagonal. Its bond is $|\langle \mu, \varepsilon_m \rangle| = 1$. Every cover has the form $\mu < s_\beta \mu$ with the dimension rising by one, so the list is complete. The Bruhat order is the transitive closure of its covers, and adding boxes one at a time (along any linear extension of $D(\kappa_2) \setminus D(\kappa_1)$) generates inclusion of shifted diagrams. □

### 3.3. **LS paths of shape $j\omega_m$.**

**Lemma 3.2.** *Let $j \geq 1$. A pair $(\tau_1 \supsetneq \cdots \supsetneq \tau_s\,;\, 0 < a_1 < \cdots < a_{s-1} < 1)$ of shifted diagrams and rationals is an LS path of shape $j\omega_m$ iff, for every $i \leq s - 1$, $2ja_i \in \mathbb{Z}$, and if $2ja_i$ is odd, then $D(\tau_i) \setminus D(\tau_{i+1})$ contains no diagonal box.*

*Proof.* Since $W_{j\omega_m} = W_{\omega_m}$, the poset is unchanged, and all bonds are multiplied by $j$. So the $a$-chain condition reads $a\,j\,b \in \mathbb{Z}$ for the bond $b \in \{1, 2\}$ of each cover in the chain. If an $a$-chain exists, it has at least one cover, so $2ja \in \mathbb{Z}$. If $ja \in \mathbb{Z}$, then every saturated chain from $\tau_{i+1}$ up to $\tau_i$ is an $a$-chain, and one exists by Lemma 3.1. If $2ja$ is odd, a cover qualifies iff its bond is 2, iff it adds an off-diagonal box. If $D(\tau_i) \setminus D(\tau_{i+1})$ has no diagonal box, every saturated chain adds only off-diagonal boxes. If it has one, every saturated chain adds it at some step. □

**Proposition 3.3** (The LS-parity model)**.** *For strict $\kappa$ with parts at most $m$ and $j \geq 0$, the LS paths of shape $j\omega_m$ with initial direction $\leq \kappa$ are in bijection with the order-preserving maps $G\colon P_\kappa \to \{0, \ldots, 2j\}$ that are even on diagonal boxes. Hence, given (G1) and (LS), $H_\kappa(j) = h^0(X_\kappa, \mathcal{O}(j)) = F_\kappa(2j)$ for all $j \geq 0$.*

*Proof.* For $j = 0$ both sides equal 1: the constant path, and $G \equiv 0$. Let $j \geq 1$ and let $\pi = (\tau_1 > \cdots > \tau_s; a)$ be an LS path with $\tau_1 \leq \kappa$, so $D(\tau_1) \subseteq D(\kappa)$ by Lemma 3.1. For $x \in D(\kappa)$ let $i(x) = \max\{i : x \in D(\tau_i)\}$ and $t(x) = a_{i(x)}$, with $t(x) = 0$ if $x \notin D(\tau_1)$. Set $g(x) = 2j\,t(x) \in \{0, \ldots, 2j\}$; this is an integer by Lemma 3.2.

- *$g$ is order-reversing.* The $D(\tau_i)$ are order ideals. If $y \leq x$ and $x \in D(\tau_i)$, then $y \in D(\tau_i)$, so $i(y) \geq i(x)$ and $g(y) \geq g(x)$.
- *$g$ is even on the diagonal.* A diagonal box $x$ with $1 \leq i(x) \leq s - 1$ lies in $D(\tau_{i(x)}) \setminus D(\tau_{i(x)+1})$, so $g(x) = 2ja_{i(x)}$ is even by Lemma 3.2. The values 0 and $2j$ are even anyway.

Conversely, let $g\colon P_\kappa \to \{0,\dots,2j\}$ be order-reversing and even on the diagonal. Let $v_1 < \cdots < v_r$ be its distinct values in the open interval $(0,2j)$. Put $a_i = v_i/2j$ and $\tau_i = \{x : g(x) \geq v_i\}$ for $i \leq r$, and $\tau_{r+1} = \{x : g(x) = 2j\}$. Each $\tau_i$ is an order ideal of $P_\kappa$, hence a shifted diagram by Lemma 2.1. They decrease strictly, because $\tau_i \setminus \tau_{i+1} = g^{-1}(v_i) \neq \varnothing$. A diagonal box in $g^{-1}(v_i)$ forces $v_i$ to be even, so Lemma 3.2 applies and we get an LS path with initial direction $\tau_1 \leq \kappa$. (If $\tau_{r+1} = \varnothing$, the last direction is the identity coset, which is allowed; if $r = 0$ the path has the single direction $g^{-1}(2j)$.) The two constructions are inverse, because $x \in \tau_i \setminus \tau_{i+1}$ iff $g(x) = v_i$ iff $t(x) = a_i$. Finally, $G = 2j - g$ is order-preserving and even on the diagonal, since $2j$ is even. The count $h^0 = \#\{\text{paths}\}$ is (G1) combined with (LS), and $h^0 = \chi$ by (G1). □

3.4. **Proof of Theorem 1.1.** Let $\Lambda \subseteq \mathbb{R}^{P_\kappa}$ be the lattice with $\mathbb{Z}$ on diagonal coordinates and $\frac{1}{2}\mathbb{Z}$ on the others. The order polytope $\mathcal{O} = \mathcal{O}(P_\kappa)$ has its vertices in $\mathbb{Z}^{P_\kappa} \subseteq \Lambda$. Rescaling the off-diagonal coordinates by 2 maps $\Lambda$ onto $\mathbb{Z}^{P_\kappa}$ and $\mathcal{O}$ onto a lattice polytope, so $L(j) = \#(j\mathcal{O} \cap \Lambda)$ is a polynomial in $j$ of degree $|\kappa|$. The map $G \mapsto y = G/2$ is a bijection from the maps counted by $F_\kappa(2j)$ onto $j\mathcal{O} \cap \Lambda$: the diagonal coordinate $y_x$ is an integer iff $G(x)$ is even. Hence $L(j) = F_\kappa(2j)$ for $j \geq 0$. (As a function of $N$, $F_\kappa(N)$ is only a quasi-polynomial of period 2; we use it only at even $N$.)

By Proposition 3.3, the polynomials $H_\kappa$ and $L$ agree at every $j \geq 0$, so $H_\kappa = L$. This proves (a) and (c). The interior of $j\mathcal{O}$ is $\{0 < y_x < j,\ y_x < y_{x'} \text{ for } x < x'\}$. Under $y = f/2$, its points in $\Lambda$ are exactly the strictly order-preserving $f\colon P_\kappa \to [2j-1]$ with even diagonal values. Ehrhart–Macdonald reciprocity for $(\mathcal{O}, \Lambda)$ gives

$$H_\kappa(-j) = L(-j) = (-1)^{|\kappa|}\#(\operatorname{int}(j\mathcal{O}) \cap \Lambda) = (-1)^{|\kappa|}N_\kappa(j) \qquad (j \geq 1),$$

which is (b). □

**Example 3.4.** For $\kappa = (1)$ the LS paths with $\tau_1 \leq (1)$ are: the constant paths $\varnothing$ and $(1)$, and the paths $(1) > \varnothing$ with time $p/2j$. The box is diagonal, so $p$ must be even, which leaves $j-1$ choices. In total there are $j+1 = H_{(1)}(j)$ paths. For $\kappa = (2)$ and $j = 1$ the paths are $\varnothing$, $(1)$, $(2)$ and $(2) > (1)$ at time $\frac{1}{2}$. The last one is allowed because the box $(0,1)$ is off-diagonal. The paths $(1) > \varnothing$ and $(2) > \varnothing$ at time $\frac{1}{2}$ are not allowed. This gives $4 = H_{(2)}(1) = (1+1)^2$.

*Remark* 3.5 (The orthogonal case). On $\mathrm{OG}(m, 2m+1)$ the weight $\omega_m$ is minuscule, so every bond is 1. The analogue of Lemma 3.2 then forces $ja_i \in \mathbb{Z}$, and the same bijection gives $H^{\mathrm{OG}}_\kappa(j) = \Omega(P_\kappa, j+1)$. We have checked by computer that all type-$B$ covers in $W/W_{\omega_m}$ have bond 1, and that the LS count equals $\Omega(P_\kappa, j+1)$ and the type-$B$ Demazure character count in 72 cases ($m \leq 4$, $j \leq 3$). We have not written out the type-$B$ analogue of Lemma 3.1, so this route is recorded as a remark only. The statement itself is classical (Remark 4.15).

*Remark* 3.6 (Computer checks). LS paths of shape $j\omega_m$ were enumerated directly from the root data, with covers, bonds and $a$-chains computed from roots rather than from the box dictionary. Their full weight multisets $\sum e^{\pi(1)}$ were compared with Demazure characters, and their number with $F_\kappa(2j)$: 164 cases ($m = 2,3$ with $j \leq 3$; $m = 4$ with $j \leq 4$; $m = 5$ with $j \leq 2$), with no failures. Two negative controls break the bond rule: setting every bond to 2, and swapping the bonds. Both fail against the Demazure characters in most cases (for example 30/32 and 28/32 failures at $m = 4$).

## 4. The $K$-theoretic proof

This section gives a second proof of Theorem 1.1(b), using (G0), (G1) and (BR) but not (LS). Its main ingredient, Theorem 4.8, is a closed formula for the $K$-theoretic Pieri coefficients of the Schubert divisor. That formula is reused in §§5–6.

4.1. **The hyperplane recursion.**

**Lemma 4.1.** *Let $X = \mathrm{LG}(m, 2m)$ or $\mathrm{OG}(m, 2m+1)$ and write*

$$[\mathcal{O}_{X_\kappa}] \cdot (1 - [L^{-1}]) = \sum_\rho c_\rho \, [\mathcal{O}_{X_\rho}] \qquad \text{in } K(X).$$

*Then $c_\rho \neq 0$ implies $\rho \subsetneq \kappa$, and for every $j \in \mathbb{C}$,*

$$H_\kappa(j) - H_\kappa(j-1) = \sum_{\rho \subsetneq \kappa} c_\rho \, H_\rho(j). \tag{1}$$

*By (G0), $c_\rho$ is the coefficient of $[\mathcal{O}_{X_\rho}]$ in $[\mathcal{O}_{X_\kappa}] \cdot \mathcal{O}^1$.*

*Proof.* Since $L$ is very ample and $X_\kappa$ is integral, there is a section $s$ of $L$ that does not vanish identically on $X_\kappa$. Let $Z \subseteq X_\kappa$ be its zero scheme. Then $[\mathcal{O}_{X_\kappa}] - [\mathcal{O}_{X_\kappa} \otimes L^{-1}] = [\mathcal{O}_Z]$. The variety $X_\kappa$ is the disjoint union of the Schubert cells $C_\rho$, $\rho \subseteq \kappa$, and $C_\kappa$ is open in $X_\kappa$. By the localization sequence $K_0(X_\kappa \setminus C_\kappa) \to K_0(X_\kappa) \to K_0(C_\kappa) \to 0$ and induction, the image of $K_0(X_\kappa)$ in $K(X)$ is spanned by the $[\mathcal{O}_{X_\rho}]$, $\rho \subseteq \kappa$. Restriction to $C_\kappa \cong \mathbb{A}^{|\kappa|}$ followed by the rank gives a map $K_0(X_\kappa) \to K_0(C_\kappa) \cong \mathbb{Z}$. It sends $[\mathcal{O}_{X_\kappa}] \mapsto 1$ and $[\mathcal{O}_{X_\rho}] \mapsto 0$ for $\rho \subsetneq \kappa$. It also sends $[\mathcal{O}_Z] \mapsto 0$, because $Z \cap C_\kappa$ is a proper closed subset of the irreducible $C_\kappa$. Hence the coefficient of $[\mathcal{O}_{X_\kappa}]$ vanishes, and $c_\rho \neq 0$ forces $\rho \subsetneq \kappa$. Multiplying by $[L^j]$ and applying $\chi$ gives (1) for integers $j$. Both sides are polynomials in $j$, so it holds for all $j$. □

4.2. **The Buch–Ravikumar Pieri rule.** We quote the statements we use from [4]. Their numbering below follows the arXiv source of [4] (§4 is OG, §5 is LG).[4]

In [4] Schubert varieties are indexed by codimension: $\mathcal{O}^\lambda = [\mathcal{O}_{X^\lambda}]$ and $\mathcal{O}^\lambda \cdot \mathcal{O}^p = \sum_\nu c^\nu_{\lambda,p} \mathcal{O}^\nu$. A shifted skew diagram $\theta$ is a *rim* if no box of $\theta$ lies strictly south and strictly east of another box of $\theta$. If $\theta$ is a nonempty rim, its *upper-right box* is the rightmost box $B$ of its top row. Its *north-east arm* $\psi$ is the largest row or column that can be obtained by intersecting $\theta$ with a square whose upper-right box is $B$. Put $\hat\theta = \theta \setminus \psi$; then $\hat\theta = \varnothing$ iff $\theta$ is a row or a column. Two boxes are *connected* if they share a side.

- [4, Cor. 5.4] For strict $\lambda \subset \nu$ with $\nu_1 \leq m$ and $0 \leq p \leq m$, $c^\nu_{\lambda,p} = \mathcal{C}(\nu/\lambda, p)$. Here $\mathcal{C}$ is an explicit alternating sum [4, eq. (11)] over subsets of the south-east corners.
- [4, Thm. 5.5] If $\theta$ is not a rim, then $\mathcal{C}(\theta, p) = 0$. If $p \leq 0$, then $\mathcal{C}(\theta, p) = \delta_{\theta,\varnothing}$. If $\theta \neq \varnothing$ is a rim, $p > 0$ and $a = |\theta| - |\hat\theta|$, then:
  - (i) if $\hat\theta = \varnothing$ and $\theta$ meets the diagonal, then $\mathcal{C}(\theta, p) = \delta_{p,|\theta|} - \delta_{p,|\theta|-1}$ if $\theta$ is a column, and $\mathcal{C}(\theta, p) = \delta_{p,|\theta|}$ otherwise;
  - (ii) if $\hat\theta = \varnothing$ and $\theta$ is disjoint from the diagonal, then $\mathcal{C}(\theta, p) = 2\delta_{p,|\theta|} - \delta_{p,|\theta|-1}$;
  - (iii) if $\hat\theta \neq \varnothing$ and $\psi$ is connected to $\hat\theta$, then $\mathcal{C}(\theta, p) = \mathcal{C}(\hat\theta, p-a) - \mathcal{C}(\hat\theta, p-a+1)$;
  - (iv) if $\hat\theta \neq \varnothing$ and $\psi$ is not connected to $\hat\theta$, then $\mathcal{C}(\theta, p) = 2\mathcal{C}(\hat\theta, p-1) - 2\mathcal{C}(\hat\theta, p)$ for $a = 1$, and $\mathcal{C}(\theta, p) = 2\mathcal{C}(\hat\theta, p-a) - 3\mathcal{C}(\hat\theta, p-a+1) + \mathcal{C}(\hat\theta, p-a+2)$ for $a > 1$.
- [4, Cor. 4.4, Lemma 4.5] (OG.) For $\lambda \subset \nu$, $c^\nu_{\lambda,p} = \mathcal{B}(\nu/\lambda, p)$. If $\theta$ is not a rim, then $\mathcal{B}(\theta, p) = 0$; if $\theta$ contains a row or column with $a$ boxes, then $\mathcal{B}(\theta, p) = 0$ for $p < a$; and if $\theta$ is a rook strip (at most one box in each row and each column), then $\mathcal{B}(\theta, 1) = (-1)^{|\theta|-1}$.

Buch and Ravikumar prove all of these. They also state a tableau formula for $c^\nu_{\lambda,p}$ on LG in terms of KLG-tableaux [4, Cor. 5.6], whose proof they leave to the reader. We do not use it: Proposition 4.6 below proves the case $p = 1$ from Thm. 5.5.

[4]The numbering is that of arXiv:1005.2605v1, in which theorems, corollaries, lemmas and propositions share one counter per section. We have not compared it with the published version. Each statement is also identified below by its content, so the citations do not depend on the numbering.

**Lemma 4.2** (Reflection)**.** *Let $R(a,b) = (m-1-b, m-1-a)$, an involution of $\Delta_m$. Then $D(\kappa^\vee) = R(\Delta_m \setminus D(\kappa))$. Consequently, for $\rho \subseteq \kappa$,*

$$D(\kappa) \setminus D(\rho) = R(D(\rho^\vee) \setminus D(\kappa^\vee)),$$

*and $\rho \subsetneq \kappa$ iff $\kappa^\vee \subsetneq \rho^\vee$. The map $R$ reverses the product order, preserves the diagonal, and exchanges rows with columns.*

*Proof.* Take $(a,b) \in \Delta_m$ and put $d = b-a+1$. Then $(a,b) \in D(\kappa)$ iff $\kappa_{a+1} \geq d$, iff $\#(\kappa \cap [d,m]) \geq a+1$. Write $R(a,b) = (\bar{a}, \bar{b})$ with $\bar{a} = m-1-b$; then $\bar{b} - \bar{a} + 1 = d$. By the same criterion, $R(a,b) \in D(\kappa^\vee)$ iff $\#(\kappa^\vee \cap [d,m]) \geq m-b$. Since $\#(\kappa \cap [d,m]) + \#(\kappa^\vee \cap [d,m]) = m-d+1 = m-b+a$, this says $\#(\kappa \cap [d,m]) \leq a$, that is, $(a,b) \notin D(\kappa)$. □

So, since $[\mathcal{O}_{X_\kappa}] = \mathcal{O}^{\kappa^\vee}$ (Lemma 2.4), the coefficient in Lemma 4.1 is $c_\rho = c^{\rho^\vee}_{\kappa^\vee,1} = \mathcal{C}(\theta_{BR}, 1)$, where $\theta_{BR} = D(\rho^\vee) \setminus D(\kappa^\vee) = R(\theta)$ and $\theta = D(\kappa) \setminus D(\rho)$.

4.3. **The Pieri coefficients count two-letter strict maps.** For a finite poset $P$ and $D \subseteq P$, let $V(P,D)$ be the number of strictly order-preserving $f\colon P \to \{e < o\}$ with $f(D) \subseteq \{e\}$. For an up-set $\theta$ of $P_\kappa$, write $V(\theta) = V(\theta, \text{diagonal boxes of } \theta)$.

**Lemma 4.3.** *$V(P,D) = 0$ if $P$ contains a chain $x < y < z$, or if some $d \in D$ is not minimal in $P$. Otherwise $V(P,D) = 2^i$, where $i$ is the number of elements of $P \setminus D$ that are comparable to no other element.*

*Proof.* A 3-chain needs three distinct values. If $x < d \in D$, then $f(x) < f(d) = e$ is impossible. Otherwise $P$ has height at most 2. A non-isolated $x$ is then either non-maximal, which forces $f(x) = e$, or non-minimal, which forces $f(x) = o$, but not both (that would give a 3-chain). These forced values satisfy every relation. A non-isolated $d \in D$ is minimal, so it is forced to $e$, consistently with $D$. An isolated element is free (two choices) unless it lies in $D$. □

In $P_\kappa$, a diagonal box $(c,c)$ of an up-set $\theta$ is non-minimal in $\theta$ iff $(c-1,c) \in \theta$. Indeed, if some $(a,b) < (c,c)$ lies in $\theta$, then $a \leq c-1$ and $b \leq c$, so $(a,b) \leq (c-1,c) < (c,c)$, and $(c-1,c) \in \theta$ because $\theta$ is an up-set.

For a shifted skew shape $\varphi$ in Buch–Ravikumar's convention, let $K(\varphi)$ be the number of labellings $\varphi \to \{1' < 1\}$ that strictly increase along every row (left to right) and every column (top to bottom) and put no $1'$ on a diagonal box, with $K(\varphi) = 0$ if $\varphi$ is not a rim.

**Lemma 4.4.** *Let $\varphi = \nu/\lambda$ be a shifted skew shape.*

*(a) If $\varphi$ is not a rim, then $\varphi$ contains a 3-chain in the product order.*
*(b) If $\varphi$ is a rim, then two boxes of $\varphi$ are comparable iff they share a row or a column.*

*Proof.* (a) Let $x = (r,c)$ and $y = (r',c')$ be boxes of $\varphi$ with $r < r'$ and $c < c'$. The right ends of the rows of a shifted strict diagram weakly decrease, so $(r,c') \in \nu$. Rows of $\lambda$ are initial segments and $(r,c) \notin \lambda$, so $(r,c') \notin \lambda$. Hence $x < (r,c') < y$ in $\varphi$. (b) Comparable means weakly north-west, and in a rim no pair is strictly north-west, so comparable boxes share a row or a column. □

**Lemma 4.5.** *For every $\rho \subsetneq \kappa$, $V(\theta) = K(R(\theta))$.*

*Proof.* If $R(\theta)$ is not a rim, it has a 3-chain by Lemma 4.4(a). Since $R$ reverses the order, $\theta$ has one too, and $V(\theta) = 0 = K(R(\theta))$. If $R(\theta)$ is a rim, then by Lemma 4.4(b) "strict along rows and columns" means "strictly order-preserving" on $R(\theta)$. Composing with $R$ (which reverses order and preserves the diagonal) and renaming $1 \mapsto e$, $1' \mapsto o$ gives a bijection. It sends strictly order-preserving maps $R(\theta) \to \{1' < 1\}$ with diagonal boxes labelled 1 to strictly order-preserving maps $\theta \to \{e < o\}$ with diagonal boxes sent to $e$. □

**Proposition 4.6.** *For every nonempty shifted skew shape $\varphi$, $\mathcal{C}(\varphi, 1) = (-1)^{|\varphi|-1}K(\varphi)$.*

This is the case $p = 1$ of [4, Cor. 5.6]. Indeed, a KLG-tableau with content $\{1\}$ uses only the labels $1' < 1$. For these labels BR's conditions (iii) and (iv), which compare a box with the boxes south-west of it, hold automatically, because 1 is the largest label and $1'$ the smallest. So the KLG-tableaux of shape $\varphi$ with content $\{1\}$ are exactly the labellings counted by $K(\varphi)$. We prove the proposition from [4, Thm. 5.5] by induction on $|\varphi|$. At $p = 1$, and using $\mathcal{C}(\hat{\varphi}, q) = 0$ for $q \leq 0$ when $\hat{\varphi} \neq \varnothing$, Thm. 5.5 reads as follows:

(i) diagonal column: 1 if $|\varphi| = 1$, $-1$ if $|\varphi| = 2$, 0 otherwise; diagonal row: 1 if $|\varphi| = 1$, 0 otherwise;
(ii) off-diagonal row or column: 2 if $|\varphi| = 1$, $-1$ if $|\varphi| = 2$, 0 otherwise;
(iii) connected arm: $-\mathcal{C}(\hat{\varphi}, 1)$ if $a = 1$, and 0 if $a \geq 2$;
(iv) disconnected arm: $-2\mathcal{C}(\hat{\varphi}, 1)$ if $a = 1$, $+\mathcal{C}(\hat{\varphi}, 1)$ if $a = 2$, and 0 if $a \geq 3$.

*Proof.* If $\varphi$ is not a rim, both sides vanish. Let $\varphi$ be a nonempty rim, and write $T(\varphi) = (-1)^{|\varphi|-1}K(\varphi)$. We use the following facts.

(g1) Rows and columns of a shifted skew shape are intervals.
(g2) A diagonal box $(c, c)$ is the leftmost box of its row and the lowest box of its column.
(g3) Let $r$ be the top row of $\varphi$ and $B = (r, c)$ its rightmost box. Then $c$ is the largest column of $\varphi$: a box $(r', c')$ with $c' > c$ would have $r' > r$, and $B$ would be strictly north-west of it. So $B$'s only possible neighbours in $\varphi$ are $L = (r, c-1)$ and $D = (r+1, c)$.
(g4) $L$ and $D$ are not both in $\varphi$, because $D$ is strictly south-east of $L$.
(g5) In a rim, comparable boxes share a row or a column (Lemma 4.4(b)), so $K$ sees only row and column constraints.
(g6) $\hat{\varphi}$ is a shifted skew shape, so the induction hypothesis applies to it. Write $\varphi = \nu/\lambda$. If $\psi = \{(r, c-a+1), \ldots, (r, c)\}$ is a row, and not ($a = 1$ and $D \in \varphi$), then $\hat{\varphi} = \nu'/\lambda$, where $\nu'$ is $\nu$ with $\psi$ removed. The set $\nu'$ is an order ideal. The only boxes that could violate this are boxes $(r+1, b) \in \nu$ with $c - a + 1 \leq b \leq c$. Such a box is not in $\lambda$, since otherwise $(r, b) \in \lambda$, so it lies in $\varphi$. For $a \geq 2$ it lies in the $a \times a$ square with upper-right box $B$, which would then not meet $\varphi$ in a row. For $a = 1$ it is $D$, which is excluded. If instead $\psi = \{(r, c), \ldots, (r+a-1, c)\}$ is a column (including $\psi = \{B\}$ with $D \in \varphi$), then $\hat{\varphi} = \nu/\lambda'$ with $\lambda' = \lambda \cup \psi$. The set $\lambda'$ is an order ideal. The box $(r-1, c)$, if it exists, lies in $\nu$ but not in $\varphi$, so it lies in $\lambda$. For $(x, c) \in \psi$, the box $(x, c-1)$, if it lies in the shifted plane, lies in $\nu$ but not in $\varphi$, so it lies in $\lambda$. For $a \geq 2$ this is because it lies in the $a \times a$ square, and for $a = 1$ it is $L$, which is excluded by (g4). In both cases Lemma 2.1 shows that the new partitions are strict.

*Cases (i), (ii): $\varphi = \psi$ is a row or a column.* With two labels, $K = 0$ for length $\geq 3$. A diagonal column has its diagonal box at the bottom (g2). Length 1 gives the label 1, so $K = 1 = T$. Length 2 gives $1'$ over 1, so $K = 1$ and $T = -1$. A diagonal row has its diagonal box leftmost. Length 1 gives $K = 1 = T$. In length 2 the leftmost box would need $1'$, which is forbidden, so $K = 0$. Off the diagonal, length 1 gives $K = 2 = T$ and length 2 gives $K = 1$ (labels $1', 1$), so $T = -1$. All of these agree with (i) and (ii).

*Cases (iii) and (iv) with $a \geq 3$.* Here $\psi$ is a row or column of at least three boxes, so $K(\varphi) = 0 = \mathcal{C}(\varphi, 1)$.

*Case (iii) with $a = 1$.* Here $\psi = \{B\}$ and $B$ is adjacent to $\hat{\varphi}$, so exactly one of $L, D$ lies in $\varphi$ (g3, g4). Let $S = (r+1, c-1)$. If $S \notin \varphi$, then the $2 \times 2$ square with upper-right box $B$ meets $\varphi$ in a row or column of two boxes, so $a \geq 2$. Hence $S \in \hat{\varphi}$.

- If $L \in \varphi$: $L$ is directly above $S$, so every valid labelling of $\hat{\varphi}$ has $L = 1'$. Then $B$, to the right of $L$, is forced to be 1, which is allowed. $B$ is alone in its column ($D \notin \varphi$; g1). The

boxes of $B$'s row other than $B$ lie in $\hat{\varphi}$, and their constraints are already part of $\hat{\varphi}$. (A box to the left of $L$ would need a label smaller than $1'$; then $K(\hat{\varphi}) = 0 = K(\varphi)$.) So $K(\varphi) = K(\hat{\varphi})$.

- If $D \in \varphi$: $S$ is directly left of $D$, so $D = 1$ in every labelling of $\hat{\varphi}$. Then $B$, above $D$, is forced to be $1'$. This is allowed, because $B$ is off the diagonal: $D = (r+1, c)$ lies in the shifted plane, so $c \geq r+1$. $B$ is alone in its row ($L \notin \varphi$; g1). The boxes of $B$'s column other than $B$ lie in $\hat{\varphi}$. (A box below $D$ would need a label larger than 1; then $K(\hat{\varphi}) = 0 = K(\varphi)$.) Again $K(\varphi) = K(\hat{\varphi})$.

Hence $T(\varphi) = -T(\hat{\varphi}) = -\mathcal{C}(\hat{\varphi}, 1)$ by induction.

*Case (iii) with $a = 2$.* Here $\psi$ is $\{L, B\}$ (a row) or $\{B, D\}$ (a column), and the $2 \times 2$ square at $B$ meets $\varphi$ in exactly $\psi$. In the row case, $S, D \notin \varphi$. The neighbours of $L$ are $(r, c-2)$, $S$, and the box above $L$, which is absent because $r$ is the top row. So the connection to $\hat{\varphi}$ goes through $(r, c-2) \in \varphi$. Then row $r$ has three boxes, and $K(\varphi) = 0$. In the column case, $S, L \notin \varphi$. The neighbours of $D$ are $(r+2, c)$, $S$ and $(r+1, c+1)$; the last is absent because column $c$ is maximal. So $(r+2, c) \in \varphi$, column $c$ has three boxes, and $K(\varphi) = 0$. Both agree with $\mathcal{C}(\varphi, 1) = 0$.

*Case (iv) with $a \in \{1, 2\}$.* Here $\psi$ is not connected to $\hat{\varphi}$.

- No box of $\psi$ shares a row or column with a box of $\hat{\varphi}$. Otherwise, walking along the common row or column interval (g1) from the $\psi$-box to the $\hat{\varphi}$-box would cross an adjacency between $\psi$ and $\hat{\varphi}$.
- $\psi$ contains no diagonal box. If $\psi$ is a row in row $r$ containing $(r, r)$, then $\psi = \{(r, r), \ldots, (r, c)\}$ by (g2). Any $(r', c') \in \hat{\varphi}$ has $r' > r$, so $r < r' \leq c' \leq c$, and column $c'$ is a column of $\psi$, a contradiction. If $\psi$ is a column in column $c$ containing $(c, c)$, then $\psi = \{(r, c), \ldots, (c, c)\}$. Any $(r', c') \in \hat{\varphi}$ has $r \leq r' \leq c' \leq c$, so row $r'$ is a row of $\psi$, a contradiction.
- By (g5) and the first bullet, the constraints on $\psi$ and on $\hat{\varphi}$ are independent, so $K(\varphi) = K(\psi)K(\hat{\varphi})$. Here $K(\psi) = 2$ for $a = 1$ (an off-diagonal single box) and $K(\psi) = 1$ for $a = 2$ (an off-diagonal domino, labels $1', 1$).
- Hence $T(\varphi) = (-1)^a K(\psi) T(\hat{\varphi})$, which is $-2\mathcal{C}(\hat{\varphi}, 1)$ for $a = 1$ and $+\mathcal{C}(\hat{\varphi}, 1)$ for $a = 2$.

Every nonempty rim falls under exactly one of the cases, since $\hat{\varphi} = \varnothing$ iff $\varphi$ is a row or a column. □

*Remark* 4.7 (Checks). The case analysis above was refereed line by line against the statement and proof of [4, Thm. 5.5]. That pass added (g6) and the parenthetical subcase of the $D$-bullet. Clio Vega independently verified the conclusion on 1243 shapes, with live negative controls, and the conclusion was also checked by machine against Buch–Ravikumar's eq. (11) [4] for all shapes arising at $m \leq 7$ (including non-rims).

**Theorem 4.8** (The Pieri coefficients)**.** *On* $\mathrm{LG}(m, 2m)$*, for every* $\rho \subsetneq \kappa$ *with* $\theta = D(\kappa) \setminus D(\rho)$*,*

$$c_\rho = (-1)^{|\theta|-1} V(\theta).$$

*Explicitly,* $c_\rho = 0$ *iff* $\theta$ *contains a 3-chain or a diagonal box* $(c, c)$ *with* $(c-1, c) \in \theta$*. Otherwise* $|c_\rho| = 2^i$*, where* $i$ *is the number of off-diagonal boxes of* $\theta$ *that are comparable to no other box of* $\theta$*.*

*Proof.* Lemma 4.2 and [4, Cor. 5.4] give $c_\rho = \mathcal{C}(R(\theta), 1)$. Proposition 4.6 turns this into $(-1)^{|\theta|-1} K(R(\theta))$, and Lemma 4.5 gives $K(R(\theta)) = V(\theta)$. The explicit form is Lemma 4.3 together with the remark following it. □

*Remark* 4.9 (Primes and odd values). Under $R$, the letters $1' < 1$ of Buch and Ravikumar become $o > e$ on $\theta$, and the rule that diagonal boxes carry no prime becomes the rule that diagonal boxes

take the even letter. In the peeling below, $e$ and $o$ are realized by consecutive values $2j < 2j+1$. So the odd values in Theorem 1.1(b) are the primed letters, and the diagonal parity condition is the "no primed diagonal" rule.

**Example 4.10.** Let $\kappa = (3,2)$, $m = 3$. Theorem 4.8 gives

$$[\mathcal{O}_{X_{(3,2)}}] \cdot \mathcal{O}^1 = 2[\mathcal{O}_{X_{(3,1)}}] - [\mathcal{O}_{X_{(2,1)}}] - [\mathcal{O}_{X_{(3)}}] + [\mathcal{O}_{X_{(2)}}].$$

The sets $\theta$ are $\{(1,2)\}$ (an isolated off-diagonal box, $V = 2$), $\{(0,2),(1,2)\}$ (a 2-chain, $V = 1$), $\{(1,1),(1,2)\}$ (diagonal box minimal, $V = 1$) and $\{(0,2),(1,1),(1,2)\}$ (height 2, $(1,1)$ minimal, $V = 1$). The signs are $(-1)^{|\theta|-1} = +,-,-,+$. The remaining candidates $\rho = (1)$ and $\rho = \varnothing$ have $c_\rho = 0$, because $\theta$ contains the 3-chain $(0,1) < (0,2) < (1,2)$. This agrees with Buch and Ravikumar's formula [4, eq. (11)] evaluated directly.

**Example 4.11** (Testing the dictionary)**.** Let $m = 3$ and $\kappa = (3,1)$, so $\kappa^\vee = (2)$. Theorem 4.8 gives $c_{(3)} = 1$ ($\theta = \{(1,1)\}$, a minimal diagonal box), $c_{(2,1)} = 2$ ($\theta = \{(0,2)\}$, an isolated off-diagonal box) and $c_{(2)} = -2$ ($\theta = \{(0,2),(1,1)\}$). The terms $\rho = (1)$ and $\rho = \varnothing$ vanish, because $(1,1)$ is not minimal in $\theta$. On Buch and Ravikumar's side, $\rho = (3)$ gives $\rho^\vee = (2,1)$ and $\theta_{BR} = (2,1)/(2) = \{(1,1)\}$, a diagonal box, so $\mathcal{C} = 1$ by [4, Thm. 5.5(i)]. And $\rho = (2,1)$ gives $\rho^\vee = (3)$ and $\theta_{BR} = \{(0,2)\}$, so $\mathcal{C} = 2$ by (ii). Now test (1) at $j = 1$ against Table 1, whose polynomials were computed from Demazure characters:

$$H_{(3,1)}(1) - H_{(3,1)}(0) = 10 - 1 = 9 = H_{(3)}(1) + 2H_{(2,1)}(1) - 2H_{(2)}(1) = 7 + 10 - 8.$$

With (3) and $(2,1)$ exchanged, the right-hand side would be $14 + 5 - 8 = 11$. A machine check, which implements [4, Thm. 5.5] and Theorem 4.8 from their printed statements, confirms (1) as an identity of polynomials for all eight $\kappa$ at $m = 3$. The exchanged dictionary fails for three of them. The shape $(2,1)$ is the quadric of Example 1.2. As posets, $P_{(3)}$ and $P_{(2,1)}$ are both three-element chains, distinguished only by the marking of the diagonal.

4.4. **The recursion and the second proof.**

**Proposition 4.12.** *On* LG*, put* $s_x(\kappa) = (-1)^{|\kappa|} H_\kappa(-x)$ *for* $x \in \mathbb{C}$*. Then for every* $x \in \mathbb{C}$*,*

$$s_{x+1}(\kappa) = s_x(\kappa) + \sum_{\rho \subsetneq \kappa} V\big(D(\kappa) \setminus D(\rho)\big)\, s_x(\rho). \tag{2}$$

*The same holds on* OG *with* $V(\theta)$ *replaced by* $|c_\rho|$*, where* $c_\rho$ *are the* OG *coefficients of Lemma 4.1.*

*Proof.* Substitute $j = -x$ in (1) and multiply by $(-1)^{|\kappa|}$. With $|\kappa| = |\rho| + |\theta|$ this gives $s_x(\kappa) - s_{x+1}(\kappa) = \sum_\rho (-1)^{|\theta|} c_\rho s_x(\rho)$. By Theorem 4.8, $(-1)^{|\theta|} c_\rho = -V(\theta)$. On OG, [4, Lemma 4.5] gives the sign $(-1)^{|\theta|-1}$ on the support (Lemma 4.14 below). □

**Lemma 4.13** (Peeling the top two values)**.** *For* $j \geq 1$ *and every* $\kappa$*,*

$$N_\kappa(j+1) = N_\kappa(j) + \sum_{\rho \subsetneq \kappa} V\big(D(\kappa) \setminus D(\rho)\big)\, N_\rho(j).$$

*Proof.* Let $f\colon P_\kappa \to [2j+1]$ be strictly order-preserving with even diagonal values, and put $\theta = f^{-1}\{2j, 2j+1\}$. If $x \in \theta$ and $x < y$, then $f(y) > f(x) \geq 2j$, so $y \in \theta$. Thus $\theta$ is an up-set, and $\theta = D(\kappa) \setminus D(\rho)$ for a unique $\rho$ (Lemma 2.1). The restriction $f|_\theta$ is strictly order-preserving into $\{2j < 2j+1\}$ and sends diagonal boxes to $2j$, the only even value. So it is one of the maps counted by $V(\theta)$, with $e = 2j$ and $o = 2j+1$. The restriction $f|_{D(\rho)}$ is counted by $N_\rho(j)$. Conversely, given $\rho$, a $V(\theta)$-map $g$ and an $N_\rho(j)$-map $h$, the union $g \cup h$ is strictly order-preserving: for $x \in D(\rho)$, $y \in \theta$ with $x < y$ we have $h(x) \leq 2j - 1 < 2j \leq g(y)$, and $y < x$ is impossible because $\theta$ is an up-set. The two constructions are inverse. The term $\rho = \kappa$ ($\theta = \varnothing$, $V = 1$) gives $N_\kappa(j)$. □

*Second proof of Theorem 1.1.* We prove (b) by induction on $j$, simultaneously for all $\kappa$. For $j = 1$, the set $[1]$ contains no even value, and every nonempty $P_\kappa$ contains the diagonal box $(0, 0)$. So $N_\kappa(1) = \delta_{\kappa,\varnothing}$, which equals $s_1(\kappa)$ by Lemma 2.6 (and $H_\varnothing \equiv 1$). For the inductive step, (2) at integer $x = j$ and Lemma 4.13 are the same recursion. Parts (a) and (c) then follow from (b) by running §3.4 backwards. Indeed, both $H_\kappa$ and $L(j) = \#(j\mathcal{O} \cap \Lambda)$ are polynomials, and by Ehrhart–Macdonald reciprocity they agree at every $j \le -1$. □

The second proof uses (G0), (G1) and (BR). It uses neither (LS) nor Ehrhart–Macdonald reciprocity for (b). The first proof uses (G1) and (LS) and not (BR). Each proof gives the equivalence of Theorem 1.1(a) and (b) by reciprocity, in opposite directions.

### 4.5. **The orthogonal Grassmannian.**

**Lemma 4.14.** *On* $\mathrm{OG}(m, 2m+1)$, $c_\rho \neq 0$ *iff* $\theta = D(\kappa) \setminus D(\rho)$ *is a nonempty set of maximal elements of* $P_\kappa$, *and then* $c_\rho = (-1)^{|\theta|-1}$.

*Proof.* By Lemma 4.2 and [4, Cor. 4.4], $c_\rho = \mathcal{B}(R(\theta), 1)$, and $R$ preserves the properties "rim" and "at most $k$ boxes per row and column". By [4, Lemma 4.5] with $p = 1 < a = 2$, $c_\rho \neq 0$ forces $R(\theta)$, hence $\theta$, to be a rook strip, and every rook strip has $c_\rho = (-1)^{|\theta|-1}$. For an up-set $\theta$ of $P_\kappa$, being a rook strip is equivalent to being an antichain. Two boxes in one row or column are comparable, so an antichain is a rook strip. Conversely, if $x = (i, j) < y = (k, l)$ in $\theta$ with $i < k$ and $j < l$, then $(i, l)$ lies in the shifted plane, below $y$ and above $x$. So $(i, l) \in \theta$ (since $\theta$ is an up-set of the order ideal $D(\kappa)$, and $(i, l) \le y$ forces $(i, l) \in D(\kappa)$), and it shares a row with $x$. An up-set is an antichain iff it consists of maximal elements, and every set of maximal elements is an up-set. □

*K-theoretic proof of Theorem 1.3.* A strictly order-preserving $f\colon P_\kappa \to [n+1]$ is determined by the set $A = f^{-1}(n+1)$, which is an up-set and an antichain, hence a set of maximal elements, together with a strictly order-preserving map $P_\kappa \setminus A \to [n]$. This gives $\overline{\Omega}(P_\kappa, n+1) = \overline{\Omega}(P_\kappa, n) + \sum_{\varnothing \neq A \subseteq \max P_\kappa} \overline{\Omega}(P_\kappa \setminus A, n)$, and $P_\kappa \setminus A = D(\rho_A)$. By Lemma 4.14, the recursion of Proposition 4.12 on OG is the same recursion with $n = j - 1$. The anchor is $\overline{\Omega}(P, 0) = \delta_{P,\varnothing} = s_1$ (Lemma 2.6). So $(-1)^{|\kappa|} H_\kappa^{\mathrm{OG}}(-j) = \overline{\Omega}(P_\kappa, j-1)$ for $j \ge 1$. Stanley reciprocity $\Omega(P, 1-j) = (-1)^{|P|}\overline{\Omega}(P, j-1)$ shows that the polynomials $H_\kappa^{\mathrm{OG}}(j)$ and $\Omega(P_\kappa, j+1)$ agree at all $j \le -1$, hence everywhere. □

*Remark* 4.15 (Theorem 1.3 is classical). Since $\omega_m$ is minuscule for $B_m$, standard monomial theory [15, 23] and the toric degeneration of Gonciulea and Lakshmibai [10] identify the homogeneous coordinate ring of $X_\kappa^{\mathrm{OG}}$ with an algebra with straightening law on the distributive lattice $J(P_\kappa)$, with the same Hilbert function as the Hibi ring [11]. Hence $H_\kappa^{\mathrm{OG}}(j) = \Omega(P_\kappa, j+1)$; compare also Proctor [21]. Stanley's reciprocity theorem then gives the negative values. We know of no single reference that states $(-1)^{|\kappa|} H_\kappa^{\mathrm{OG}}(-j) = \overline{\Omega}(P_\kappa, j-1)$ in this form, but it is a one-line composite of these results, and we claim no novelty for it.[5]

## 5. The integer window

*Proof of Theorem 1.4 from the counts.* On LG, $s_j(\kappa) = N_\kappa(j)$ by Theorem 1.1(b). On OG, $s_j(\kappa) = \overline{\Omega}(P_\kappa, j-1)$ by Theorem 1.3. Lemma 2.3 gives the vanishing range, the positivity beyond it, and the monotonicity. For $\kappa = \varnothing$, $s_j \equiv 1$ and $W = 0$. □

[5] The references in this remark ([15, 23, 10, 11, 21]) were not read first-hand for this paper. Nothing else in the paper depends on them: §4.5 proves Theorem 1.3 independently.

The theorem also has a direct proof by induction. That proof does not pass through any counting formula for $s_j$. It uses only (2), the anchor, a bound on how much $r$ can drop along the support of the Pieri rule, and one explicit nonzero Pieri term. We include it because it shows why integers are easier than half-integers: the anchor value 0 makes the recursion (2) monotone.

**Lemma 5.1** (Chain drop)**.** *If $c_\rho \neq 0$, then every chain of $P_\kappa$ meets $\theta = D(\kappa) \setminus D(\rho)$ in at most two boxes on* LG*, and in at most one box on* OG*. Hence $r(\rho) \geq r(\kappa) - 2$ on* LG*, and $r(\rho) \geq r(\kappa) - 1$ on* OG*. The same bounds hold for $r(\rho')$ and $r(\kappa')$.*

*Proof.* On LG, $\theta$ has no 3-chain by Theorem 4.8 and Lemma 4.3. On OG, $\theta$ is an antichain by Lemma 4.14. If $C$ is a longest chain of $P_\kappa$, then $C \setminus \theta \subseteq D(\rho)$ is a chain of the claimed size. For the primed version, take $C$ inside the rows $\geq 1$, which form copies of $P_{\kappa'}$ and $P_{\rho'}$ (§2.1). Row 0 is the largest part of both $\kappa$ and $\rho$. □

**Lemma 5.2** (A term that drops the window)**.** *Let $\kappa \neq \varnothing$, $r = r(\kappa)$, and for $t \in \{1, 2\}$ put $T_t = \{(a, b) \in D(\kappa) : a + b \geq r - t\}$ and $D(\rho_t) = D(\kappa) \setminus T_t$.*

*(i) $\rho_t$ is a strict partition with $\rho_t \subsetneq \kappa$. If $t \leq r$, then $r(\rho_t) \leq r - t$.*
*(ii) On* LG*, let $t^* = 1$ if $r$ is odd and $t^* = 2$ if $r$ is even. Then $c_{\rho_{t^*}} \neq 0$.*
*(iii) On* OG*, $c_{\rho_1} = (-1)^{|T_1|-1}$.*
*(iv) $W(\rho_{t^*}) \leq W(\kappa) - 1$ on* LG*, and $W(\rho_1) \leq W(\kappa) - 1$ on* OG*.*

*Proof.* (i) $T_t$ is an up-set of $P_\kappa$, so $\rho_t$ exists by Lemma 2.1. It contains the box realising $\max(a+b)$ (Lemma 2.2), so $\rho_t \neq \kappa$. If $t \leq r$ and $\rho_t \neq \varnothing$, every box of $D(\rho_t)$ has $a + b \leq r - t - 1$, so $r(\rho_t) \leq r - t$ by Lemma 2.2. If $\rho_t = \varnothing$, then $r(\rho_t) = 0 \leq r - t$. (The hypothesis $t \leq r$ is needed: for $\kappa = (1)$ and $t = 2$ the bound would read $0 \leq -1$. We only use $t = t^* \leq r$.)

(ii) By Theorem 4.8 and Lemma 4.3 it suffices to show that $T_{t^*}$ has no 3-chain and that each of its diagonal boxes is minimal in it. The coordinate sum strictly increases along a chain, and $T_{t^*}$ lies on at most two antidiagonals $a + b = \text{const}$, so there is no 3-chain. A box of $T_{t^*}$ strictly below $(c, c)$ has a smaller coordinate sum. For $r$ odd, $T_1$ is a single antidiagonal, so no such box exists. For $r$ even, $T_2 = \{a + b \in \{r - 2, r - 1\}\}$, and $2c$ is even, so $2c = r - 2$ lies on the inner antidiagonal; again nothing in $T_2$ lies below $(c, c)$.

(iii) $T_1$ is the top antidiagonal: it is nonempty, and it consists of maximal elements of $P_\kappa$. Apply Lemma 4.14.

(iv) On LG with $r$ odd, $r(\rho_1) \leq r - 1$, which is even, so $\lceil r(\rho_1)/2 \rceil \leq (r - 1)/2 = W - 1$. With $r$ even, $r(\rho_2) \leq r - 2$, so $\lceil r(\rho_2)/2 \rceil \leq r/2 - 1 = W - 1$. On OG, $r(\rho_1) \leq r - 1 = W - 1$. □

The parity choice in (ii) matters. For $r$ odd the two-strip $T_2$ has its diagonal box (if any) on the *outer* antidiagonal, where it can lie above $(c - 1, c) \in T_2$; then $V(T_2) = 0$. A machine check of the 67 odd-$r$ two-strips at $m \leq 6$ against Buch–Ravikumar's formula finds that 42 of them have vanishing coefficient, each one exactly in this situation.

*Inductive proof of Theorem 1.4.* We prove by strong induction on $|\kappa|$ that $s_j(\kappa) \geq 0$ for all integers $j \geq 1$, with equality iff $j \leq W(\kappa)$. For $\kappa = \varnothing$ this is clear. Let $\kappa \neq \varnothing$.

*Monotonicity.* For $j \geq 1$, every $\rho$ in (2) has $|\rho| < |\kappa|$, so $s_j(\rho) \geq 0$ by induction, and the coefficients in (2) are nonnegative. Hence $s_{j+1}(\kappa) \geq s_j(\kappa)$.

*Anchor.* $s_1(\kappa) = 0$ by Lemma 2.6. So $s_j(\kappa) \geq 0$ for all $j \geq 1$.

*Vanishing.* Let $1 \leq j \leq W - 1$. For every $\rho$ with $c_\rho \neq 0$, Lemma 5.1 gives $W(\rho) \geq W - 1 \geq j$: on LG, $\lceil r(\rho)/2 \rceil \geq \lceil (r - 2)/2 \rceil = W - 1$, and on OG, $r(\rho) \geq r - 1$. So $s_j(\rho) = 0$ by induction, and (2) gives $s_{j+1}(\kappa) = s_j(\kappa)$. Starting from $s_1 = 0$, this reaches $s_W = 0$.

*Nonvanishing.* At $j = W$, (2) gives $s_{W+1}(\kappa) \geq |c_{\rho^*}|\, s_W(\rho^*)$ with $\rho^* = \rho_{t^*}$ (on LG) or $\rho_1$ (on OG); the other terms are $\geq 0$. By Lemma 5.2, $c_{\rho^*} \neq 0$ and $W(\rho^*) < W$, so $s_W(\rho^*) > 0$ by induction. Monotonicity gives $s_j > 0$ for all $j > W$. □

**Example 5.3.** For $\kappa = (3,2)$ on $\mathrm{LG}(3,6)$: the coordinate sums of $D(\kappa)$ are $0,1,2,2,3$, so $r = 4$ and $W = 2$. By Example 4.10, (2) at $j = 2$ reads $s_3(3,2) = s_2(3,2) + 2s_2(3,1) + s_2(2,1) + s_2(3) + s_2(2)$. The first four terms vanish, since each of those $\kappa$ has $W = 2$, and $s_2(2) = 1$. So $s_3(3,2) = 1$, that is, $H_{(3,2)}(-3) = -1$. Here $\rho^* = \rho_2 = (2)$ and $T_2 = \{(0,2),(1,1),(1,2)\}$. At $j = 3$ the same recursion gives $s_4(3,2) = 1 + 2\cdot 2 + 1 + 3 + 4 = 13$. For $\kappa = (4,3,2,1)$ we have $r = 7$ and $W = 4$; the support of $[\mathcal{O}_{X_\kappa}]\cdot\mathcal{O}^1$ is the single term $\rho = (4,3,2)$ with $\theta = T_1 = \{(3,3)\}$ and $c_\rho = 1$.

## 6. Half-integer values

Throughout this section $X = \mathrm{LG}(m,2m)$, and $s_x(\kappa) = (-1)^{|\kappa|}H_\kappa(-x)$ for $x \in \frac{1}{2}\mathbb{Z}$. Since (2) is an identity of polynomials, it holds at half-integers as well. Write $y_i = e^{-\varepsilon_i}$.

### 6.1. The anchor at $-\frac{1}{2}$ and the bridge to vector bundles.

For $1 \le k \le m$, put $c(k) = s_{m-k+1}s_{m-k+2}\cdots s_m$, of length $k$. Then $D_{c(k)} = D_{m-k+1}\circ\cdots\circ D_m$, so $D_m$ acts first.

**Lemma 6.1** (Reduced word)**.** *$w_\kappa = c(\kappa_\ell)\cdots c(\kappa_2)\,c(\kappa_1)$, and this word is reduced. Equivalently, $w_\kappa = w_{\kappa'}\,c(\kappa_1)$ with lengths adding, and every letter of $w_{\kappa'}$ is $s_j$ with $j \ge m - \kappa_1 + 2$.*

*Proof.* Let $p = m - \kappa_1 + 1$. Applying the letters of $c(\kappa_1)$ to $\omega = (1,\dots,1)$ from right to left, $s_m$ flips the last coordinate and $s_{m-1},\dots,s_p$ carry the $-1$ to position $p$. So $c(\kappa_1)\omega = v_{(\kappa_1)}$. Every letter of $c(\kappa_\ell)\cdots c(\kappa_2)$ is $s_j$ with $j \ge m - \kappa_2 + 1 \ge p + 1$, so it only moves coordinates $\ge p+1$, where $v_{(\kappa_1)}$ is $+1$. By induction on $\ell(\kappa)$, the word maps $\omega$ to $v_\kappa$. It has length $|\kappa| = \ell(w_\kappa)$ and lies in the coset $w_\kappa W_P$, and the minimal coset representative is the unique element of minimal length. So the word is reduced and equals $w_\kappa$. □

Let $D^B_m g = (g - y_m s_m g)/(1 - y_m)$ be the type-$B$ operator for the short root $\varepsilon_m$. Put $D'_i = D_i$ for $i < m$ and $D'_m = (1+y_m)^{-1}D^B_m$, and let $\Phi = \prod_{i=1}^m (1+y_i)^{-1}$, an analytic germ with $\Phi(0) = 2^{-m}$.

**Lemma 6.2** (Conjugation)**.** *For all $i$, $D_i(e^{-\omega/2}g) = e^{-\omega/2}D'_i(g)$ and $D'_i(\Phi g) = \Phi D_i(g)$. Hence $D_w(e^{-\omega/2}g) = e^{-\omega/2}\,\Phi\,D_w(\Phi^{-1}g)$ for every $w$.*

*Proof.* For $i < m$, both $e^{-\omega/2}$ and $\Phi$ are $s_i$-invariant, and $D_i(\varphi g) = \varphi D_i g$ whenever $s_i\varphi = \varphi$. For $i = m$: $s_m(e^{-\omega/2}) = e^{-\omega/2}e^{\varepsilon_m}$ and $e^{-\alpha_m} = y_m^2$, so $D_m(e^{-\omega/2}g) = e^{-\omega/2}(g - y_m s_m g)/(1-y_m^2) = e^{-\omega/2}(1+y_m)^{-1}D^B_m g$. For the second identity, write $\Phi = \Phi'(1+y_m)^{-1}$ with $\Phi'$ $s_m$-invariant, and use $s_m(1+y_m)^{-1} = y_m(1+y_m)^{-1}$. Then $D^B_m((1+y_m)^{-1}g) = (g - y_m^2 s_m g)/((1+y_m)(1-y_m))$, and multiplying by $(1+y_m)^{-1}\Phi'$ gives $\Phi(g - y_m^2 s_m g)/(1-y_m^2) = \Phi D_m g$. □

**Proposition 6.3** (Bridge)**.** *For every integer $a \ge 0$, $2^m H_\kappa(-a - \frac{1}{2}) = \big[D_{w_\kappa}(F_a)\big]_{y=1}$, where $F_a = \prod_{i=1}^m y_i^a(1+y_i)$.*

*Proof.* By Lemma 2.5,

$$H_\kappa(-a-\tfrac{1}{2}) = \big[D_{w_\kappa}(e^{-\omega/2}e^{-a\omega})\big](0).$$

By Lemma 6.2 this equals $[e^{-\omega/2}\Phi\,D_{w_\kappa}(F_a)](0)$, because $\Phi^{-1}e^{-a\omega} = F_a$, and $e^{-\omega/2}\Phi = 2^{-m}$ at 0. □

*Remark* 6.4. $F_a$ is the character of $\mathbb{C}_{-a\omega}\otimes\bigwedge^\bullet U_0$, where $U_0$ is the $P$-module with weights $-\varepsilon_i$, the fibre of the tautological subbundle $U$. Pulling back to $G/B$ and filtering by $B$-weights, (G1) turns Proposition 6.3 into $2^m H_\kappa(-a-\frac{1}{2}) = \chi(X_\kappa, \bigwedge^\bullet U \otimes \mathcal{O}(-a))$. So the half-integer values are Euler characteristics of honest vector bundles. (For $m = 1$, $X = \mathbb{P}^1$ and $U = \mathcal{O}(-1)$: $\chi(\mathcal{O}\oplus\mathcal{O}(-1)) = 1 = 2H(-\frac{1}{2})$.) None of the proofs below use this geometric reading.

**Theorem 6.5.** *$H_\kappa(-\frac{1}{2}) = 2^{-\kappa_1}$ for every $\kappa$ (with $\kappa_1 = 0$ for $\kappa = \varnothing$).*

*Proof.* By Proposition 6.3 with $a = 0$, $2^m H_\kappa(-\frac{1}{2}) = [D_{w_\kappa} \prod_i (1+y_i)]_{y=1}$. Let $p = m - \kappa_1 + 1$. We claim that $D_{c(\kappa_1)} \prod_{i=1}^m (1+y_i) = \prod_{i<p}(1+y_i)$. Apply $D_m, D_{m-1}, \ldots, D_p$ in turn. Before $D_q$ acts, the function is $\prod_{i\le q}(1+y_i)$ (true for $q = m$), and $\prod_{i<q}(1+y_i)$ is $s_q$-invariant. So $D_q \prod_{i\le q}(1+y_i) = \prod_{i<q}(1+y_i)\,(1 + D_q e^{-\varepsilon_q})$. Here $\langle -\varepsilon_q, \alpha_q^\vee\rangle = -1$ for $q < m$ and for $q = m$, so $D_q e^{-\varepsilon_q} = 0$. By Lemma 6.1, the remaining letters $s_j$ have $j \ge p+1$; they fix $\prod_{i<p}(1+y_i)$, and $D_j(1) = 1$. Evaluating at $y = 1$ gives $2^{p-1} = 2^{m-\kappa_1}$. □

**Theorem 6.6.** *If $\kappa_1 \ge 2$, then $H_\kappa(-\frac{3}{2}) = 2^{-\kappa_1} H_{\kappa'}(-1)$, and $H_{(1)}(-\frac{3}{2}) = -\frac{1}{2}$. Hence $H_\kappa(-\frac{3}{2}) = 0$ iff $\ell(\kappa) \ge 2$.*

*Proof.* For $\kappa = (1)$, $H_{(1)}(j) = j+1$. Let $\kappa_1 \ge 2$, $p = m-\kappa_1+1$, and $f(y) = y+y^2$, so $F_1 = \prod_i f(y_i)$. We apply $D_{c(\kappa_1)}$. $D_m$ kills $y_m$ (pairing $-1$) and sends $y_m^2$ to $-1$ (pairing $-2$), so $D_m F_1 = -\prod_{i<m} f(y_i)$. Next, $D_{m-1}$ kills $y_{m-1}$ and sends $y_{m-1}^2$ to $-y_{m-1}y_m$, so $D_{m-1}(-f(y_{m-1})) = y_{m-1}y_m$. For $p \le q < m-1$, $D_q$ applied to $f(y_q)y_{q+1}\cdots y_m$ fixes $y_q y_{q+1}\cdots y_m$ (pairing 0) and kills $y_q^2 y_{q+1}\cdots y_m$ (pairing $-1$). Hence $D_{c(\kappa_1)}F_1 = \prod_{i<p} f(y_i)\cdot y_p \cdot (y_{p+1}\cdots y_m)$. By Lemma 6.1, $D_{w_{\kappa'}}$ acts only on the coordinates $p+1, \ldots, m$, where $y_{p+1}\cdots y_m = e^{-\omega''}$ for the rank-$(\kappa_1-1)$ block. The prefactor is inert and equals $2^{p-1}$ at $y = 1$. The Demazure expression for $H_{\kappa'}$ does not depend on the rank, by the same inert-factor argument. Hence $2^m H_\kappa(-\frac{3}{2}) = 2^{m-\kappa_1} H_{\kappa'}(-1)$. Finally, $H_{\kappa'}(-1) = 0$ iff $\kappa' \ne \varnothing$ (Lemma 2.6), and $H_\varnothing \equiv 1$. □

6.2. **Vanishing by hyperplane sections.**

**Theorem 6.7.** *For every strict $\kappa$ and every odd $M$ with $3 \le M \le 2\lceil r(\kappa')/2\rceil + 1$, $H_\kappa(-M/2) = 0$.*

*Proof.* Strong induction on $|\kappa|$. If $\ell(\kappa) \le 1$, then $r(\kappa') = 0$ and the range is empty. Let $\ell(\kappa) \ge 2$, so $r' = r(\kappa') \ge 1$. Put $b_\kappa(x) = H_\kappa(x) - H_\kappa(x-1)$.

*Every term vanishes on the shrunken window.* Let $M$ be odd with $3 \le M \le 2\lceil r'/2\rceil - 1$, and let $c_\rho \ne 0$. By Lemma 5.1, $r(\rho') \ge r' - 2$, so $2\lceil r(\rho')/2\rceil + 1 \ge 2\lceil (r'-2)/2\rceil + 1 = 2\lceil r'/2\rceil - 1 \ge M$. Thus $M$ lies in the window of $\rho$, and $H_\rho(-M/2) = 0$ by induction. By (1), $b_\kappa(-M/2) = 0$.

*Telescoping from the anchor.* By Theorem 6.6, $H_\kappa(-\frac{3}{2}) = 0$. For $M$ in the range above, $b_\kappa(-M/2) = 0$ says $H_\kappa(-M/2) = H_\kappa(-(M+2)/2)$. Going up $M = 3, 5, \ldots, 2\lceil r'/2\rceil - 1$ carries the zero from $-\frac{3}{2}$ to $-(2\lceil r'/2\rceil + 1)/2$. □

Theorem 6.7 uses only the support of the Pieri rule: it needs to know that $c_\rho \ne 0$ implies that $\theta$ has no 3-chain. This also follows from the rim condition of [4, Thm. 5.5] together with the bound of two boxes per row and column (two boxes in strictly south-east position, or three in a row or column, would be needed for a 3-chain). The coefficients do not cooperate at the edge of the window: the nonzero terms of $b_\kappa$ at $-M^*/2$, $M^* = 2\lceil r'/2\rceil + 1$, need not have a common sign (for $\kappa = (3,2)$ they are $-\frac{1}{8}$ and $+\frac{1}{4}$). So the nonvanishing half needs the finer argument below.

6.3. **The odd-zero law.** Let $A_o$ and $A_e$ be the matrices indexed by strict partitions with $A_o(\kappa,\nu) = [\theta \text{ is an antichain without diagonal boxes}]$ and $A_e(\kappa,\nu) = [\theta \text{ is an antichain}]$, for $\nu \subseteq \kappa$ and $\theta = D(\kappa) \setminus D(\nu)$, and 0 otherwise.

**Lemma 6.8** (Layers). *(a) $(A_o A_e)(\kappa,\nu) = V(D(\kappa)\setminus D(\nu))$. Hence (2) reads $s_{x+1} = A_o A_e\, s_x$ for every $x \in \mathbb{C}$.*

*(b) For $a \ge 1$, $[(A_o A_e)^{a-1} A_o](\kappa,\nu) = N_{\kappa/\nu}(a)$.*

*Proof.* Peel values from the top, as in Lemma 4.13. The preimage of the current top value is an antichain that is an up-set of what remains. The value is odd (no diagonal boxes allowed: $A_o$) or even ($A_e$). In (b), the values $2a-1, 2a-2, \ldots, 1$ alternate $o, e, \ldots, o$, giving $A_o(A_eA_o)^{a-1} = (A_oA_e)^{a-1}A_o$. Part (a) is the same with the alphabet $\{e < o\}$, and $\theta = \varnothing$ gives 1. □

**Lemma 6.9** (The anchor collapses to single rows)**.** *Let $\varepsilon = A_e\, s_{1/2}$. Then $\varepsilon(\varnothing) = 1$, $\varepsilon((L)) = (-1)^{L+1}2^{-L}$ for $L \geq 1$, and $\varepsilon(\kappa) = 0$ if $\ell(\kappa) \geq 2$.*

*Proof.* By Theorem 6.5, $s_{1/2}(\nu) = (-1)^{|\nu|}2^{-\nu_1}$, and $\varepsilon(\kappa) = \sum_{A \subseteq \max P_\kappa} s_{1/2}(\kappa \setminus A)$. If $\ell(\kappa) \geq 2$, the last box $z$ of the bottom row is maximal and not in row 0. Toggling $z \in A$ changes $|\nu|$ by one and leaves $\nu_1$ alone, so the terms cancel in pairs. If $\kappa = (L)$, the only maximal box is $(0, L-1)$, and $\varepsilon = (-1)^L 2^{-L} + (-1)^{L-1}2^{-(L-1)} = (-1)^{L+1}2^{-L}$. □

**Corollary 6.10** (Row formula)**.** *For $a \geq 1$, $s_{a+1/2}(\kappa) = N_\kappa(a) + \sum_{L=1}^{\kappa_1}(-1)^{L+1}2^{-L}N_{\kappa/(L)}(a)$.*

*Proof.* $s_{a+1/2} = (A_oA_e)^a s_{1/2} = (A_oA_e)^{a-1}A_o\,\varepsilon$; apply Lemmas 6.8(b) and 6.9. □

For example, for $\kappa = (3)$ and $a = 2$: $s_{5/2}(3) = 0 + \frac{1}{2}\cdot 3 - \frac{1}{4}\cdot 3 + \frac{1}{8}\cdot 1 = \frac{7}{8}$, and indeed $H_{(3)}(-\frac{5}{2}) = -\frac{7}{8}$.

*The fibre over the lower rows.* Let $n = \kappa_1$, $M = 2a-1$, and $k = \kappa_2$ (with $k = 0$ if $\ell(\kappa) = 1$). Rows $\geq 1$ of $P_\kappa$ form a copy of $P_{\kappa'}$. Row 0 is the chain $(0,0) < \cdots < (0, n-1)$, and its cover relations to rows $\geq 1$ are $(0,\beta) < (1,\beta)$ for $1 \leq \beta \leq k$; nothing in rows $\geq 1$ lies below a box of row 0. Fix a strictly order-preserving $g$ on rows $\geq 1$ with values in $[M]$ and even diagonal values; there are $N_{\kappa'}(a)$ of them. Put $u_\beta = g(1,\beta)$ for $1 \leq \beta \leq k$ and $u_\beta = M+1$ (no bound) for $k < \beta \leq n-1$. For a bound vector $u$ and $1 \leq L \leq n$, let $G_L(M;u)$ be the number of strictly increasing $v_L < \cdots < v_{n-1}$ in $[M]$ with $v_\beta < u_\beta$ (so $G_n = 1$), and let $E_0(M;u)$ be the same count for $L = 0$ with $v_0$ even. Put

$$\Phi(u) = E_0(M;u) - \sum_{L=1}^{n}(-\tfrac{1}{2})^L G_L(M;u).$$

Summing Corollary 6.10 over $g$ gives

$$s_{a+1/2}(\kappa) = \sum_g \Phi(u(g)). \tag{3}$$

The bounds coming from an actual $g$ satisfy

$$u_\beta \geq \beta + 1 \quad \text{for } 1 \leq \beta \leq k, \tag{H}$$

because $u_1 = g(1,1)$ is even, hence at least 2, and row 1 increases strictly.

Let $\Psi_L(M;u) = \sum_{L'=L}^{n}(-\frac{1}{2})^{L'-L}G_{L'}(M;u)$. Then $\Psi_n = 1$ and $\Phi = E_0 + \frac{1}{2}\Psi_1$.

**Lemma 6.11** (Shift)**.** *Let $1 \leq L \leq n-1$ and $M \geq 1$, and suppose $u_{L'} \geq 2$ for $L \leq L' \leq n-1$. Then $\Psi_L(M;u) = G_L(M-1;u-1) + \frac{1}{2}\Psi_{L+1}(M-1;u-1)$.*

*Proof.* For $L' \geq L$, let $\alpha_{L'}$ (resp. $\beta_{L'}$) count the fillings of boxes $L', \ldots, n-1$ with $v_{L'} = 1$ (resp. $v_{L'} \geq 2$), and put $\beta_n = 1$. Then $G_{L'} = \alpha_{L'} + \beta_{L'}$ for $L' \leq n-1$. Deleting box $L'$ (value 1) is a bijection onto the fillings of $L'+1, \ldots, n-1$ with all values $\geq 2$. Its inverse puts 1 in box $L'$, which is allowed because $u_{L'} \geq 2$ and $M \geq 1$. So $\alpha_{L'} = \beta_{L'+1}$ for $L \leq L' \leq n-1$. Substituting,

$$\Psi_L = \sum_{L'=L}^{n-1}(-\tfrac{1}{2})^{L'-L}(\beta_{L'} + \beta_{L'+1}) + (-\tfrac{1}{2})^{n-L}\beta_n = \beta_L + \tfrac{1}{2}\sum_{L''>L}(-\tfrac{1}{2})^{L''-L-1}\beta_{L''}.$$

Finally, $v \mapsto v-1$ identifies fillings with values in $[2, M]$ under bounds $u$ with fillings in $[M-1]$ under bounds $u-1$, so $\beta_{L''}(M;u) = G_{L''}(M-1;u-1)$, including $L'' = n$. □

**Lemma 6.12** (One row)**.** *Assume (H).*

*(a) If $M \geq n-1$, then $\Phi(u) = E_0(M;u) + \sum_{i=0}^{n-2} 2^{-i-1}G_{1+i}(M-1-i;u-1-i) + 2^{-n} \;\geq\; 2^{-n}$.*

*(b) If $M \leq n-1$, then $\Phi(u) = (-1)^{n-1-M}2^{-n}$.*

*Proof.* Apply Lemma 6.11 repeatedly. Step $i = 0, 1, \ldots$ rewrites $\Psi_{1+i}(M-i; u-i)$. Its hypotheses are $M - i \geq 1$ and $u_{L'} - i \geq 2$ for $L' \geq 1 + i$. The second holds by (H) for bounded boxes ($u_{L'} - i \geq L' + 1 - i \geq 2$), and for unbounded boxes as long as $M - i \geq 1$. (a) If $M \geq n-1$, steps $i = 0, \ldots, n-2$ are allowed, ending at $\Psi_n = 1$, so $\Psi_1 = \sum_{i=0}^{n-2} 2^{-i} G_{1+i}(M-1-i; u-1-i) + 2^{-(n-1)}$. Every term is nonnegative. (b) If $M \leq n-1$, steps $i = 0, \ldots, M-1$ are allowed, ending at $\Psi_{1+M}(0; u-M) = (-\frac{1}{2})^{n-1-M}$, since only $G_n = 1$ survives an empty alphabet. The $G$-terms vanish: $G_{1+i}(M-1-i)$ fills $n-1-i$ boxes from $M-1-i < n-1-i$ values. So $\Psi_1 = 2^{-M}(-\frac{1}{2})^{n-1-M}$. Also $E_0 = 0$, because $n$ strictly increasing values that are all $\geq 2$ need $M \geq n+1$. Hence $\Phi = \frac{1}{2}\Psi_1$. □

*Proof of Theorem 1.5.* The value at $-\frac{1}{2}$ is Theorem 6.5. Let $a \geq 1$ and $M = 2a - 1$. If $2a \leq \kappa_1$, then $M \leq n-1$, and Lemma 6.12(b) gives $\Phi \equiv (-1)^{n-2a} 2^{-n} = (-1)^{\kappa_1} 2^{-\kappa_1}$ for every $g$. So $s_{a+1/2}(\kappa) = (-1)^{\kappa_1} 2^{-\kappa_1} N_{\kappa'}(a)$ by (3). If $2a \geq \kappa_1$, then $M \geq n-1$, and Lemma 6.12(a) gives $\Phi \geq 2^{-n}$ for every $g$, so $s_{a+1/2}(\kappa) \geq 2^{-\kappa_1} N_{\kappa'}(a)$. (For $2a = \kappa_1$ both apply, and they agree because $\kappa_1$ is even.) In both cases $s_{a+1/2}(\kappa) = 0$ iff $N_{\kappa'}(a) = 0$. By Lemma 2.3, this happens iff $\kappa' \neq \varnothing$ and $a \leq \lceil r(\kappa')/2 \rceil$. The signs follow from $H_\kappa = (-1)^{|\kappa|} s$. Each $\Phi(u)$ lies in $2^{-n}\mathbb{Z}$, so $2^{\kappa_1} s_{a+1/2}(\kappa) \in \mathbb{Z}$. □

**Example 6.13.** Take $\kappa = (3, 2)$ and $a = 2$, so $M = 3 \geq n - 1 = 2$. On rows $\geq 1$, the chain $(1,1) < (1,2)$ with $(1,1)$ even and values in $[3]$ admits only $g = (2, 3)$, so $u = (2, 3)$. Then $E_0 = 0$ (no $v_1 < 2$ fits above an even $v_0$) and $G_1(2; (1, 2)) = 0$. Lemma 6.12(a) gives $\Phi = \frac{1}{8}$, so $s_{5/2}(3, 2) = \frac{1}{8} = 2^{-\kappa_1} N_{(2)}(2)$, and $H_{(3,2)}(-\frac{5}{2}) = -\frac{1}{8}$. The bound in Theorem 1.5(iii) is attained.

*Remark* 6.14 (Checks). The values $(-1)^{|\kappa|} H_\kappa(-a - \frac{1}{2})$, computed from type-$C$ Demazure characters and interpolated to $x = -a - \frac{1}{2}$, agree with $\sum_g \Phi(u(g))$ for all $\kappa$ with parts at most $m \leq 4$ and $a \leq 5$ (125 cases). Lemma 6.12 holds pointwise in all 477 cases that occur. Hypothesis (H) is necessary: without it $\Phi$ can be negative (for example $u = (1)$ gives $-\frac{1}{4}$).

6.4. **Nonvanishing for large half-integers.**

*Proof of Corollary 1.6.* Let $M = 2a + 1 \geq 2m + 1$, so $a \geq m \geq \kappa_1$ and $2a \geq \kappa_1$. The parts of $\kappa'$ are at most $m-1$, so $\max(a' + b') \leq 2m - 4$ on $D(\kappa')$, and $r(\kappa') \leq 2m - 3$ by Lemma 2.2. Hence $\lceil r(\kappa')/2 \rceil \leq m - 1 < a$, and Theorem 1.5 gives $s_{a+1/2}(\kappa) \geq 2^{-\kappa_1} N_{\kappa'}(a) > 0$ with $2^{\kappa_1} s_{a+1/2}(\kappa) \in \mathbb{Z}$. Then $2^m s_{a+1/2}(\kappa)$ is an integer, and it is at least $2^{m-\kappa_1} \geq 1$. □

*Remark* 6.15 (A cohomological formula). Corollary 1.6 also follows from a positive cohomological formula. That argument uses more geometry: normality and the Cohen–Macaulay property of Schubert varieties, Kodaira-type vanishing via Frobenius splitting, and the formula $\omega_Y = \mathcal{O}_Y(-\partial Y) \otimes L_{-\rho}$ for Schubert varieties $Y \subseteq G/B$ [3, Ch. 3]. Let $\pi\colon Y \to X_\kappa$ be the $P/B$-bundle. Serre duality applied to Remark 6.4, followed by a $B$-stable filtration of $\pi^*(\omega_{X_\kappa} \otimes \bigwedge^\bullet U^\vee(a))$, gives, for $a \geq m$,

$$(-1)^{|\kappa|} 2^m H_\kappa(-a - \tfrac{1}{2}) = \sum_{S \subseteq [m]} h^0(Y, \mathcal{O}_Y(-\partial Y) \otimes L_{\lambda_S}), \qquad \lambda_S = (a - i + [i \in S])_{i=1}^m.$$

Every $\lambda_S$ is dominant for $a \geq m$, and the term $S = [m]$ is nonzero. We record this as a remark because the proof given above needs only (G0), (G1) and (BR).

6.5. **The half-step operator.** By Lemma 6.8(a), $s_{x+1} = A_o A_e\, s_x$. In this subsection all matrices are indexed by strict partitions with parts at most $m$ and vanish unless $\nu \subseteq \kappa$. Such matrices form an algebra, and a matrix in it with 1's on the diagonal (unitriangular) has at most one unitriangular square root: if $U^2 = V$, then $V(\kappa, \nu) = 2U(\kappa, \nu) + \sum_{\nu \subsetneq \rho \subsetneq \kappa} U(\kappa, \rho) U(\rho, \nu)$ determines $U(\kappa, \nu)$ by induction on $|\kappa| - |\nu|$. Let $T$ be the unitriangular square root of $A_o A_e$.

**Proposition 6.16.** *(a)* $s_{x+1/2} = T\, s_x$ *for every* $x \in \mathbb{C}$.

*(b) Each entry* $T(\kappa,\nu)$ *is the product of weights of the connected components of* $\theta = D(\kappa) \setminus D(\nu)$*, where a component* $D$ *has weight:* $1$ *if* $D$ *is a single off-diagonal box;* $0$ *if* $D$ *has no diagonal box and at least two boxes;* $0$ *if* $D$ *contains at least two diagonal boxes;* $0$ *if* $D$ *contains exactly one diagonal box and contains a* $2\times 2$ *square; and* $\varepsilon\,(-\frac12)^{|D|}$ *if* $D$ *is a ribbon (contains no* $2\times 2$ *square) with exactly one diagonal box, where* $\varepsilon = +1$ *if the left neighbour of the north-east end box of* $D$ *lies in* $D$*, and* $\varepsilon = -1$ *otherwise. In particular every entry is* $0$ *or* $\pm 2^{-k}$.

For example, a single diagonal box has weight $\frac12$, and $T((3),\varnothing) = -\frac18$. At $x = 1$, Lemma 2.6 gives $s_1 = \delta_\varnothing$, so by (a) the $\varnothing$-column $T(\cdot,\varnothing)$ is $s_{3/2}$, and (b) evaluates it: $\theta = D(\kappa)$ has $\ell(\kappa)$ diagonal boxes in one component, so $T(\kappa,\varnothing) = 0$ if $\ell(\kappa) \ge 2$, while $T((k),\varnothing) = (-\frac12)^k$ for $k \ge 2$ and $T((1),\varnothing) = \frac12$. This is Theorem 6.6, whose factor $2^{-\kappa_1}$ is thus the cost of deleting the row $\kappa_1$; the $2^{-\kappa_1}$ of Theorem 6.5 is instead the column $Ts_0 = S\,P(\cdot,\varnothing)$ at $x = 0$ in the proof of (a). A natural guess, that $T$ is given by a shifted Murnaghan–Nakayama rule, is false: the two agree only on single boxes, and their supports differ.

The proof rests on two explicit matrices. Write $\kappa' = (\kappa_2,\kappa_3,\dots)$, set $\kappa_r = 0$ for $r > \ell(\kappa)$, and use the conventions $\nu_1 = 0$ and $\nu' = \varnothing$ for $\nu = \varnothing$. For $\nu \ne \kappa$ let $r_0 = \min\{r : \nu_r \ne \kappa_r\}$, and put

$$[\nu \prec \kappa] = [\nu_s \le \kappa_{s+1} \text{ for all } s \ge r_0].$$

In words, $\nu$ keeps the first $r_0 - 1$ parts of $\kappa$, and its remaining parts fit under $(\kappa_{r_0+1},\kappa_{r_0+2},\dots)$; this forces $\nu \subseteq \kappa$ and $\ell(\nu) < \ell(\kappa)$. Define

$$X(\kappa,\nu) = \delta_{\kappa\nu} - [\nu \ne \kappa]\,[\ell(\nu) = \ell(\kappa) - 1]\,2^{-\kappa_{r_0}}[\nu \prec \kappa],$$
$$P(\kappa,\nu) = \delta_{\kappa\nu} + [\nu \ne \kappa]\,2^{-\kappa_{r_0}}[\nu \prec \kappa].$$

Let $\zeta(\kappa,\nu) = [\nu \subseteq \kappa]$, let $\zeta_o(\kappa,\nu) = [\nu \subseteq \kappa,\ \ell(\nu) = \ell(\kappa)]$, and let $S$ be the diagonal matrix with entries $(-1)^{|\kappa|}$.

**Lemma 6.17** (Möbius functions)**.** $A_e = S\zeta^{-1}S$ *and* $A_o = S\zeta_o^{-1}S$.

*Proof.* Fix $\nu \subseteq \kappa$. By Lemma 2.1, $A_e(\kappa,\rho) \ne 0$ means that $A = D(\kappa) \setminus D(\rho)$ is an antichain and an up-set, that is, a set of maximal boxes of $P_\kappa$; and $\nu \subseteq \rho$ means $A \subseteq \theta$. Since $\theta$ is an up-set, its maximal boxes are maximal in $P_\kappa$. Hence

$$(SA_eS\,\zeta)(\kappa,\nu) = \sum_{A \subseteq \max\theta} (-1)^{|A|} = [\max\theta = \varnothing] = \delta_{\kappa\nu}.$$

For $A_o$, the sets $A$ are sets of off-diagonal maximal boxes, and $\zeta_o(\rho,\nu)$ adds the condition $\ell(\nu) = \ell(\rho) = \ell(\kappa)$. The diagonal boxes of $\theta$ are the $(i,i)$ with $\ell(\nu) \le i < \ell(\kappa)$, so then $\theta$ has no diagonal box, and the same computation gives $(SA_oS\,\zeta_o)(\kappa,\nu) = [\ell(\nu) = \ell(\kappa)][\theta = \varnothing] = \delta_{\kappa\nu}$. □

**Lemma 6.18.** *(a) For* $\kappa \ne \varnothing$ *and all* $\nu$*,* $X(\kappa,\nu) = [\nu_1 = \kappa_1]X(\kappa',\nu') - 2^{-\kappa_1}\zeta_o(\kappa',\nu)$ *and* $P(\kappa,\nu) = [\nu_1 = \kappa_1]P(\kappa',\nu') + 2^{-\kappa_1}\zeta(\kappa',\nu)$.

*(b) If* $\nu_1 \le \rho_1$*, then* $P(\rho,\nu') = 2^{-\rho_1}[\nu \subseteq \rho]$.

*(c)* $X\zeta = \zeta_o P$.

*(d)* $XP = I$.

*Proof.* (a) If $\nu_1 = \kappa_1$, then $r_0$ and all conditions shift by one when the first parts are removed, and $\zeta_o(\kappa',\nu) = \zeta(\kappa',\nu) = 0$ because $\nu_1 > \kappa_2$. If $\nu_1 \ne \kappa_1$, then $r_0 = 1$ and $[\nu \prec \kappa] = [\nu \subseteq \kappa']$.

(b) If $\rho = \varnothing$, then $\nu = \nu' = \varnothing$ and both sides are 1. Otherwise $\nu'_1 < \rho_1$, so $\nu' \ne \rho$ and $r_0 = 1$, and $P(\rho,\nu') = 2^{-\rho_1}[\nu_{s+1} \le \rho_{s+1} \text{ for } s \ge 1]$. Together with $\nu_1 \le \rho_1$ the bracket is $[\nu \subseteq \rho]$.

(c) Induction on $\ell(\kappa)$; for $\kappa = \varnothing$ both sides are $\delta_{\varnothing\nu}$. Let $\ell = \ell(\kappa) \geq 1$. By (a), and since $[\nu \subseteq (\kappa_1, \rho')] = [\nu_1 \leq \kappa_1][\nu' \subseteq \rho']$,

$$(X\zeta)(\kappa, \nu) = [\nu_1 \leq \kappa_1](X\zeta)(\kappa', \nu') - 2^{-\kappa_1}(\zeta_o\zeta)(\kappa', \nu).$$

On the other side, $\rho \subseteq \kappa$ with $\ell(\rho) = \ell$ means $\rho = (a, \rho')$ with $\zeta_o(\kappa', \rho') = 1$ and $\rho'_1 < a \leq \kappa_1$. By (a) and $\sum_{a=\rho'_1+1}^{\kappa_1} 2^{-a} = 2^{-\rho'_1} - 2^{-\kappa_1}$,

$$(\zeta_o P)(\kappa, \nu) = \sum_{\rho'} [\rho'_1 < \nu_1 \leq \kappa_1] P(\rho', \nu') + \sum_{\rho'} 2^{-\rho'_1} \zeta(\rho', \nu) - 2^{-\kappa_1}(\zeta_o\zeta)(\kappa', \nu),$$

with $\rho'$ running over $\zeta_o(\kappa', \rho') = 1$. By induction, $(X\zeta)(\kappa', \nu') = \sum_{\rho'} P(\rho', \nu')$. So it remains to show $[\nu_1 \leq \kappa_1] \sum_{\rho'} [\rho'_1 \geq \nu_1] P(\rho', \nu') = \sum_{\rho'} 2^{-\rho'_1} [\nu \subseteq \rho']$. If $\nu_1 > \kappa_1 > \kappa_2$, both sides vanish. Otherwise the two sums agree term by term, by (b) when $\rho'_1 \geq \nu_1$, and because both terms vanish when $\rho'_1 < \nu_1$.

(d) Induction on $\ell(\kappa)$, trivial for $\kappa = \varnothing$. Expanding $X(\kappa, \rho)$ by (a), and then $P(\rho, \nu)$ for $\rho = (\kappa_1, \rho')$ by (a),

$$(XP)(\kappa, \nu) = [\nu_1 = \kappa_1](XP)(\kappa', \nu') + 2^{-\kappa_1}\big((X\zeta)(\kappa', \nu) - (\zeta_o P)(\kappa', \nu)\big).$$

By induction and (c), this is $[\nu_1 = \kappa_1]\delta_{\kappa'\nu'} = \delta_{\kappa\nu}$. □

*Proof of Proposition 6.16.* Put $\tilde{T} = \zeta_o^{-1} X$. By Lemma 6.18(c), $\tilde{T} = P\zeta^{-1}$. By Lemma 6.18(d),

$$\tilde{T}^2 = \zeta_o^{-1}(XP)\zeta^{-1} = \zeta_o^{-1}\zeta^{-1}.$$

By Lemma 6.17, $(S\tilde{T}S)^2 = A_o A_e$, and $S\tilde{T}S$ is unitriangular, so

$$T = S\zeta_o^{-1} X S = A_o\,(SXS), \qquad \tilde{T}\zeta = P.$$

(a) Lemma 2.6 gives $s_1 = \delta_\varnothing$, so $s_0 = (A_o A_e)^{-1}\delta_\varnothing = S\zeta\zeta_o S\,\delta_\varnothing$, that is, $s_0(\kappa) = (-1)^{|\kappa|}$. Hence $Ts_0 = S\tilde{T}\mathbf{1} = S\,(\tilde{T}\zeta)(\cdot, \varnothing) = S\,P(\cdot, \varnothing)$, whose $\kappa$-entry is $(-1)^{|\kappa|}2^{-\kappa_1} = s_{1/2}(\kappa)$ by Theorem 6.5. Since $T$ commutes with $T^2 = A_o A_e$, for integers $j \geq 0$ we get $s_{j+1/2} = (A_o A_e)^j T s_0 = T(A_o A_e)^j s_0 = T s_j$. Both sides of $s_{x+1/2} = T s_x$ are polynomial in $x$ entrywise, so they agree for all $x$.

(b) We have $T(\kappa, \nu) = \sum_\rho A_o(\kappa, \rho)(-1)^{|\rho|-|\nu|} X(\rho, \nu)$. Here $A_o$ preserves the length, and $X$ either fixes $\nu$ or lowers the length by exactly one.

If $\ell(\nu) = \ell(\kappa)$, then $T(\kappa, \nu) = A_o(\kappa, \nu)$. The set $\theta$ is convex in $\Delta_m$, so two comparable boxes of $\theta$ are joined by a path of adjacent boxes of $\theta$. Hence $\theta$ is an antichain iff all its components are single boxes, which are off-diagonal. If $\ell(\nu) \leq \ell(\kappa) - 2$, then $T(\kappa, \nu) = 0$, and the diagonal boxes $(i, i)$, $\ell(\nu) \leq i < \ell(\kappa)$, lie in one component together with the boxes $(i, i+1)$, so the product of weights is 0 as well.

Let $\ell(\nu) = \ell(\kappa) - 1 = \ell - 1$. Give the box $(a, b)$ the content $b - a$. Row $r$ of $\rho/\nu$ (rows numbered from 1) is the content interval $[\nu_r, \rho_r - 1]$, and a box of content $c$ in row $r$ lies directly above the box of content $c - 1$ in row $r + 1$. So two nonempty rows $r, r+1$ are adjacent iff $\nu_r \leq \rho_{r+1}$. A skew shape is connected iff its nonempty rows are consecutive and successive rows are adjacent, and its contents then form an interval. Since row $\ell(\rho)$ is nonempty and starts at content 0, for $\ell(\nu) = \ell(\rho) - 1$ we get

$$X(\rho, \nu) = -\,2^{-(1+c(\rho/\nu))}\,[\rho/\nu \text{ is connected}],$$

where $c(\cdot)$ is the largest content; indeed $r_0$ is the first nonempty row, its largest content is $\rho_{r_0} - 1$, and $[\nu \prec \rho]$ is the connectivity condition. The term $A_o(\kappa, \rho)$ removes a set $A$ of off-diagonal maximal boxes of $\theta$. Hence

$$T(\kappa, \nu) = -\sum_A (-1)^{|\theta|-|A|}\,2^{-(1+c(\theta\setminus A))}\,[\theta \setminus A \text{ is connected}].$$

Let $D_0$ be the component of $\theta$ containing its unique diagonal box $d$. If $\theta \setminus A$ is connected, $A$ contains every other component, so these are single boxes (two adjacent boxes are comparable). Hence $T(\kappa,\nu) = 0$ unless every other component is a single box, and then $T(\kappa,\nu) = w(D_0)$, where

$$w(D) = -\sum_B (-1)^{|D|-|B|}\, 2^{-(1+c(D\setminus B))},$$

the sum running over sets $B$ of off-diagonal maximal boxes of $D$ with $D \setminus B$ connected.

Let $D$ occupy rows $r_0, \dots, \ell$ with content intervals $[a_r, b_r]$. Both $a$ and $b$ strictly decrease, $a_\ell = 0$, and $a_r \le b_{r+1} + 1$. The maximal boxes are row ends, and the end of row $r$ is maximal iff $r = \ell$ or $b_{r+1} \le b_r - 2$. Moreover $D$ contains a $2\times 2$ square iff $a_{r-1} \le b_r$ for some $r_0 < r \le \ell$.

*$D$ contains a $2\times 2$ square.* Let $r^*$ be the largest $r$ with $a_{r-1} \le b_r$, and let $x$ be the end of row $r^*$. If $x$ were not maximal, then $b_{r^*+1} = b_{r^*} - 1$ and $a_{r^*} < a_{r^*-1} \le b_{r^*}$ would give $a_{r^*} \le b_{r^*+1}$, contradicting the choice of $r^*$. Also $x \ne d$, since its content is at least $a_{r^*-1} \ge 1$. Toggling $x$ in $B$ preserves connectivity: row $r^*$ keeps a box, rows $r^*-1$ and $r^*$ stay adjacent because $a_{r^*-1} \le b_{r^*}$, row $r^*-1$ has at least two boxes and so is never emptied, and the adjacency of rows $r^*$ and $r^*+1$ does not involve $b_{r^*}$. Toggling $x$ also preserves the largest content, which is at least $b_{r^*-1} - 1 \ge b_{r^*}$. The sign changes, so $w(D) = 0$.

*$D$ is a ribbon*, that is, $a_{r-1} = b_r + 1$ for $r_0 < r \le \ell$. Then $D$ has one box of each content, so $|D| = b_{r_0} + 1$. Removing the end of a row $r > r_0$ disconnects rows $r-1$ and $r$, unless all rows above $r$ are emptied. That would require removing the north-east end box $e$ from a top row $\{e\}$, but then $a_{r_0} = b_{r_0} = b_{r_0+1} + 1$, so $e$ lies above a box of $D$ and is not maximal. Hence $B \subseteq \{e\}$. The box $e$ is an off-diagonal maximal box iff row $r_0$ has at least two boxes, that is, iff $\varepsilon = +1$, and then $D \setminus \{e\}$ is a connected ribbon with largest content $b_{r_0} - 1$. Therefore

$$w(D) = -(-\tfrac12)^{|D|} + [\varepsilon = +1]\cdot 2(-\tfrac12)^{|D|} = \varepsilon\,(-\tfrac12)^{|D|}. \qquad \square$$

In words: $T = A_o(I + SQS)$ with $Q = X - I$. The half-step removes an off-diagonal antichain and then deletes one part $\kappa_{r_0}$, at cost $2^{-\kappa_{r_0}}$, shrinking the parts below it. The identities of Lemma 6.18 and Proposition 6.16 were also checked by computer for all strict partitions with parts at most 8, and $T$ was compared with the binomial series for $(A_oA_e)^{1/2}$ for parts at most 6.

## 7. Examples

7.1. **The quadric $\mathrm{LG}(2,4)$ and $\mathbb{P}^3 = \mathrm{OG}(2,5)$.** Let $\kappa = (2,1)$, so $P_\kappa$ is the chain $(0,0) < (0,1) < (1,1)$, with both ends diagonal.

*Positive degrees.* For $j = 1$, the order-preserving maps $g\colon P_\kappa \to \{0,1,2\}$ that are even at both ends are $(0,0,0)$, $(0,0,2)$, $(0,1,2)$, $(0,2,2)$ and $(2,2,2)$. This gives $H(1) = 5 = \dim V_{\mathrm{Sp}_4}(\omega_2)$, the number of Plücker coordinates of the quadric $\mathrm{LG}(2,4) \subseteq \mathbb{P}^4$.

*Negative integers.* $H(j) = (j+1)(j+2)(2j+3)/6$ gives $H(-1) = H(-2) = 0$, $H(-3) = -1$ and $H(-4) = -5$, matching Example 1.2. Here $r = 3$ and $W = 2$. By Serre duality with $\omega_{Q^3} = \mathcal{O}(-3)$, $H(-3) = -h^0(\mathcal{O})$.

*Half-integers.* $H(-\frac12) = \frac14 = 2^{-\kappa_1}$, and $H(-\frac32) = 0$ because $\ell(\kappa) = 2$ (Theorem 6.6). Next, $H(-\frac52) = -\frac14$: here $\kappa' = (1)$ and $r(\kappa') = 1$, so the half-integer window is $a \le 1$. Since $2a = 4 \ge \kappa_1$, the sign is $(-1)^{|\kappa|} = -1$, and $|H(-\frac52)| = 2^{-2}N_{(1)}(2) = \frac14$.

*The orthogonal twin.* For $\mathrm{OG}(2,5) \cong \mathbb{P}^3$ with the spinor bundle $\mathcal{O}_{\mathbb{P}^3}(1)$, $H^{\mathrm{OG}}(j) = \Omega(P_{(2,1)}, j+1) = \binom{j+3}{3}$. Its zeros are $-1,-2,-3$, so $W = r = 3$, and $H^{\mathrm{OG}}(-4) = -1 = -\overline{\Omega}(P_{(2,1)}, 3)$. The window on OG is twice as long, as Theorem 1.4 predicts.

7.2. **All Schubert varieties of** $\mathrm{LG}(3,6)$**.** Table 1 lists $H_\kappa$ for all $\kappa \neq \varnothing$ with parts at most 3, together with $r = r(\kappa)$ and $r' = r(\kappa')$. Table 2 lists the values at $j = -1, \dots, -5$ and at $j = -\frac{1}{2}, \dots, -\frac{9}{2}$. The polynomials were computed from Demazure characters (G1) by interpolation, independently of the theorems. The integer values agree with the strict counts of Theorem 1.1(b) for $j \leq 4$. The zeros are exactly those predicted by Theorems 1.4 and 1.5: integer zeros $-1, \dots, -\lceil r/2 \rceil$, and half-integer zeros $-a - \frac{1}{2}$ for $1 \leq a \leq \lceil r'/2 \rceil$ when $\kappa' \neq \varnothing$.

| $\kappa$ | $r$ | $r'$ | $H_\kappa(j)$ |
|---|---|---|---|
| $(1)$ | 1 | 0 | $j+1$ |
| $(2)$ | 2 | 0 | $(j+1)^2$ |
| $(3)$ | 3 | 0 | $(j+1)(j+2)(4j+3)/6$ |
| $(2,1)$ | 3 | 1 | $(j+1)(j+2)(2j+3)/6$ |
| $(3,1)$ | 3 | 1 | $(j+1)^2(j+2)(2j+3)/6$ |
| $(3,2)$ | 4 | 2 | $(j+1)(j+2)(2j+3)(2j^2+6j+5)/30$ |
| $(3,2,1)$ | 5 | 3 | $(j+1)(j+2)^2(j+3)(2j+3)(2j+5)/180$ |

TABLE 1. Hilbert polynomials of the Schubert varieties of $\mathrm{LG}(3,6)$.

| $\kappa$ | $-1$ | $-2$ | $-3$ | $-4$ | $-5$ | $-\frac{1}{2}$ | $-\frac{3}{2}$ | $-\frac{5}{2}$ | $-\frac{7}{2}$ | $-\frac{9}{2}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| $(1)$ | 0 | $-1$ | $-2$ | $-3$ | $-4$ | $\frac{1}{2}$ | $-\frac{1}{2}$ | $-\frac{3}{2}$ | $-\frac{5}{2}$ | $-\frac{7}{2}$ |
| $(2)$ | 0 | 1 | 4 | 9 | 16 | $\frac{1}{4}$ | $\frac{1}{4}$ | $\frac{9}{4}$ | $\frac{25}{4}$ | $\frac{49}{4}$ |
| $(3)$ | 0 | 0 | $-3$ | $-13$ | $-34$ | $\frac{1}{8}$ | $\frac{1}{8}$ | $-\frac{7}{8}$ | $-\frac{55}{8}$ | $-\frac{175}{8}$ |
| $(2,1)$ | 0 | 0 | $-1$ | $-5$ | $-14$ | $\frac{1}{4}$ | 0 | $-\frac{1}{4}$ | $-\frac{5}{2}$ | $-\frac{35}{4}$ |
| $(3,1)$ | 0 | 0 | 2 | 15 | 56 | $\frac{1}{8}$ | 0 | $\frac{3}{8}$ | $\frac{25}{4}$ | $\frac{245}{8}$ |
| $(3,2)$ | 0 | 0 | $-1$ | $-13$ | $-70$ | $\frac{1}{8}$ | 0 | $-\frac{1}{8}$ | $-\frac{17}{4}$ | $-\frac{259}{8}$ |
| $(3,2,1)$ | 0 | 0 | 0 | 1 | 14 | $\frac{1}{8}$ | 0 | 0 | $\frac{1}{8}$ | $\frac{35}{8}$ |

TABLE 2. Values of $H_\kappa$ at negative integers and half-integers on $\mathrm{LG}(3,6)$.

A few entries, read through the theorems:

- $(3)$: $s_3(3) = 3$ counts the triples $2 \leq x < y < z \leq 5$ with $x$ even, namely $x = 2$ and $\{y, z\} \subseteq \{3, 4, 5\}$. At $-\frac{3}{2}$ we have $2a = 2 \leq \kappa_1 = 3$, so $H = 2^{-3} N_\varnothing = \frac{1}{8}$ with sign $(-1)^{|\kappa| - \kappa_1} = +$.
- $(3,1)$: $H(-\frac{5}{2}) = \frac{3}{8} > 0$. Here $2a = 4 \geq 3$, so the sign is $(-1)^{|\kappa|} = +$, and the lower bound $2^{-3} N_{(1)}(2) = \frac{1}{8}$ is not attained.
- $(3,2,1) = \mathrm{LG}(3,6)$: $r = 5$, so the integer zeros are $-1, -2, -3$. Since $\kappa' = (2,1)$ has $r' = 3$, the half-integer zeros are $-\frac{3}{2}$ and $-\frac{5}{2}$. Then $H(-4) = 1$ and $H(-\frac{7}{2}) = \frac{1}{8} = 2^{-3} N_{(2,1)}(3)$.

7.3. **Computer checks.** Apart from the checks recorded with each section, Theorem 1.1(b) and Theorem 1.3 were compared with Hilbert polynomials computed directly from Demazure characters at negative weights, for all $\kappa$ with parts at most $m$: $16, 32, 64, 128$ pairs $(\kappa, j)$ with $j \leq 4$ for $m = 2, 3, 4, 5$, on both LG and OG, with no failures. Negative controls fail as they should (alphabet $[2j]$ instead of $[2j-1]$; no diagonal parity; OG alphabet $[j]$). An independent census by Clio Vega, who evaluated the Demazure characters directly at negative weights, confirmed Theorem 1.4 for all strict $\kappa$ with parts at most 6, on both LG and OG.

## 8. Open questions

**Question 8.1** (Other cominuscule spaces)**.** The cominuscule $G/P$ that are not minuscule are $\mathrm{LG}(m, 2m)$ and the odd quadrics $Q^{2m-1} = B_m/P_1$. For the odd quadric the Bruhat poset is a chain, and exactly one cover has bond 2. This is the lecture-hall setting of [6]. Is the Hilbert function of every Schubert variety in a cominuscule $G/P$ the Ehrhart polynomial of its order polytope with respect to the lattice determined by the bonds, as in Theorem 1.1(c)? Is there a uniform reason, independent of the case-by-case bond computation of Lemma 3.1?

**Question 8.2** (Canonical modules)**.** For $j \geq 1$, Kodaira-type vanishing on Schubert varieties gives $(-1)^{|\kappa|} H_\kappa(-j) = h^0(X_\kappa, \omega_{X_\kappa}(j))$. Is there a standard-monomial-type basis of $H^0(X_\kappa, \omega_{X_\kappa}(j))$ indexed by the strictly order-preserving maps of Theorem 1.1(b)? Which $X_\kappa$ are Gorenstein, in terms of the interior of $(\mathcal{O}(P_\kappa), \Lambda)$?

**Question 8.3** (Orthogonal half-integers)**.** On $\mathrm{OG}(m, 2m+1)$ the half-integer values of $H^{\mathrm{OG}}_\kappa$ have sporadic zeros, for example $H^{\mathrm{OG}}_{(3,2)}(-\frac{5}{2}) = 0$ and $H^{\mathrm{OG}}_{(5,1)}(-\frac{3}{2}) = 0$. We found ten such $\kappa$ at $m = 6$, and no pattern. Is there an OG analogue of Theorem 1.5?

**Question 8.4** (Semi-toric degenerations)**.** Chirivì's semi-toric degeneration of $X_\kappa$ [5], as summarised in [6, §7], has one toric component for each maximal chain $\varnothing < \cdots < \kappa$, that is, for each standard shifted tableau of shape $\kappa$. The bond vector of such a chain lies in $\{1, 2\}^{|\kappa|}$, with 1 exactly at diagonal boxes. The associated lecture hall polytope appears to be the simplex of Stanley's canonical triangulation of $\mathcal{O}(P_\kappa)$ that corresponds to the chain, taken in the lattice $\Lambda$. Is Chirivì's degeneration a geometric realisation of this triangulation of $(\mathcal{O}(P_\kappa), \Lambda)$? We have not compared it with Chirivì's gluing data.

**Question 8.5** (Higher Pieri coefficients)**.** Proposition 4.6 is the case $p = 1$ of the KLG-tableau formula [4, Cor. 5.6], whose proof is left to the reader there. A complete proof for all $p$, by the same induction over Theorem 5.5 of [4], would be useful. Do the coefficients $c^{\nu}_{\lambda,p}$ admit an interpretation by strict maps into a $2p$-letter alphabet $\{e_1 < o_1 < \cdots < e_p < o_p\}$ with even letters on the diagonal, extending Theorem 4.8?

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